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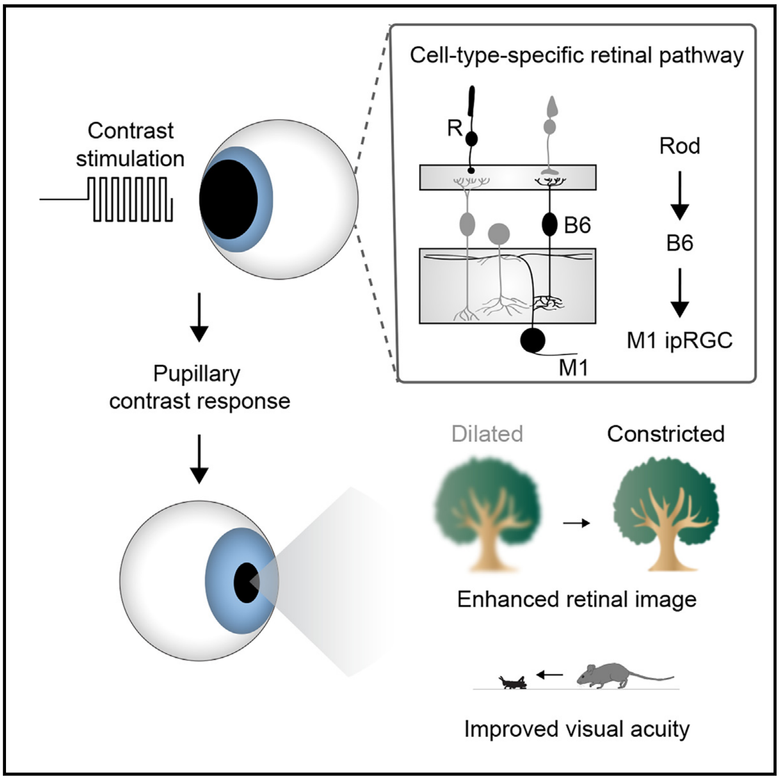
A pupillary contrast response in mice and humans: Neural mechanisms and visual functions

Graphical abstract



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In brief

The pupil is well known to constrict in response to increases in luminance. Here, Fitzpatrick et al. show that temporal contrast also drives pupil construction via a cell-type-specific retinal circuit in mice and humans. The pupillary contrast response enhances high spatial frequency contrast in the retinal image and improves visual acuity.

Highlights

- Contrast elicits robust pupil constriction in mice and humans
- A cell-type-specific retinal pathway mediates the pupillary contrast response
- Pupil size shapes spatial frequency content in the retinal image
- Pupil constriction enhances contrast sensitivity and visual acuity in mice

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A pupillary contrast response in mice and humans: Neural mechanisms and visual functions

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SUMMARY

In the pupillary light response (PLR), increases in ambient light constrict the pupil to dampen increases in retinal illuminance. Here, we report that the pupillary reflex arc implements a second input-output transformation; it senses temporal contrast to enhance spatial contrast in the retinal image and increase visual acuity. The pupillary contrast response (PCoR) is driven by rod photoreceptors via type 6 bipolar cells and M1 ganglion cells. Temporal contrast is transformed into sustained pupil constriction by the M1's conversion of excitatory input into spike output. Computational modeling explains how the PCoR shapes retinal images. Pupil constriction improves acuity in gaze stabilization and predation in mice. Humans exhibit a PCoR with similar tuning properties to mice, which interacts with eye movements to optimize the statistics of the visual input for retinal encoding. Thus, we uncover a conserved component of active vision, its cell-type-specific pathway, computational mechanisms, and optical and behavioral significance.

INTRODUCTION

Animals, including humans, do not encounter sensory input passively but actively seek salient information and shape its content through motor actions (i.e., active sensing).[1–3] In vision, saccadic head and eye movements shift the gaze to regions of interest.[4–7] Fixational eye movements prevent visual fading between saccades and alter the frequency composition of the retinal input.[8,9] A second component of active vision is the pupil; all visual information passes through the pupil before it is processed by the retina. In the pupillary light response (PLR), rising ambient light levels constrict the pupil to dampen increases in retinal illuminance.[10,11] Pupil constriction also enhances the spatial contrast of retinal images, reducing blur caused by peripheral lens aberrations.[12–14] Whether the pupillary reflex arc senses contrast to guide its adjustments, if so, which circuits and mechanisms detect contrast and convert it into pupil constriction, and how the two components of active vision (i.e., head/eye movements and the pupil) cooperate to shape the retinal image content are unknown.

The PLR in mice and humans is driven by melanopsin-expressing intrinsically photosensitive retinal ganglion cells (ipRGCs).[15–18] Six ipRGC types can be distinguished by morphology, transcriptomics, and function in mice (M1–M6)[19–24] and two in primates (M1 and M2).[25–29] The PLR depends on M1 ipRGCs projecting to the olivary pretectal nucleus (OPN).[16] In addition to intrinsic photoresponses from melanopsin, M1 ipRGCs receive synaptic excitation conveying rod and, to a lesser extent, cone signals.[22,28,30] How this excitation reaches M1 ipRGCs remains mysterious.[31–33] In the mouse retina, 15 bipolar cell types relay photoreceptor signals from the outer to the inner retina.[34–37] Nine respond to light increments (i.e., ON bipolar cells) and stratify their axons in the inner part of the retina's inner plexiform layer (IPL); six respond to light decrements (i.e., OFF bipolar cells) and stratify their axons in the outer part of the IPL.[34–37] M1 ipRGC dendrites target the outer IPL but receive input from ON bipolar cell axons passing through.[22,31–33] Anatomical studies showed that type 6 ON bipolar cells form en passant synapses with M1 ipRGCs.[31–33] The function of these unusual connections remains to be tested.

M1 ipRGCs encode luminance over a wide range of light levels and timescales.[22,28,29,38–40] In luminance encoding, synaptic and intrinsic excitation cover distinct, overlapping stimulus domains. Synaptic excitation drives fast responses and extends coverage to dim-light stimuli, whereas intrinsic excitation supports sustained responses, particularly at high light levels.[28,30,39] This division of labor is evident in the PLR's sensitivity to manipulations of intrinsic and synaptic M1 ipRGC excitation.[17,41–43] Bipolar cells, which provide synaptic excitation to M1 ipRGCs, encode contrast (i.e., fluctuations around the mean light level) more than luminance (i.e., the mean light level).[33,35,37,44] It is unclear whether M1 ipRGCs also encode contrast and, if so, to what end.

The retina supports a wide range of visual behaviors.[45–49] Visual behaviors are thought to rely on either image-forming (i.e., contrast encoding)[50–52] or non-image-forming (i.e., luminance encoding) functions of the retina.[53,54] The discovery of luminance signals in brain areas associated with image-forming functions and contrast signals in brain areas mediating non-image-forming functions[30,55–58] blurs these categorical boundaries and suggests that retinal luminance and contrast information may converge to optimize performance in some visual behaviors.

Here, we discover a pupillary contrast response (PCoR) in mice and humans. We identify a cell-type-specific pathway from rod photoreceptors to M1 ipRGCs that conveys contrast information and decipher how M1 ipRGCs encode contrast and luminance to drive pupil constriction. We use computational modeling and experimental perturbations to reveal the optical and behavioral significance of the PCoR and examine its interaction with the PLR and the cooperation of pupil and gaze dynamics in shaping the retinal input.

RESULTS

Contrast elicits robust pupil constriction in mice

The PLR has been characterized in detail,[10,41] but whether the pupil responds to contrast is unclear. We measured the consensual pupil response in awake head-fixed mice to address this question. We presented stimuli from a green LED, illuminating the eye contralateral to the recorded eye (Figure 1A).[50] Light steps of increasing brightness elicited increasing pupil constriction (Figure 1B, $EC_{50} = 10^{2.17 \pm 0.34}$ R*), as expected.[10,41] To our surprise, temporal fluctuations around a stimulus mean (5 Hz, 50% contrast) drove further constriction of light-adapted pupils ($10^{3.5}$ R*, Figures 1C and S1). This response exceeded the response to a 50%-contrast light step without temporal modulation, even though the average light intensity of the step stimulus was 1.5-fold greater than that of the modulated stimulus (Figure 1D). Indeed, a step stimulus required a 10-fold higher average light intensity to drive a comparable pupil constriction to the temporally modulated stimulus (Figure S1). Thus, the pupil responds robustly and specifically to temporal contrast (i.e., the PCoR). We next varied the temporal frequency and modulation depth of the stimulus. The PCoR exhibited band-pass temporal tuning (Figure 1C), with strong responses from 0.5 to 10 Hz, and increased in proportion to the logarithm of the stimulus contrast (Figure 1E). At frequencies $\leq$ 2 Hz, the pupil tracked the stimulus frequency (i.e., F1 response, Figure S2) in addition to its sustained constriction (i.e., F0 response).

To measure pupil responses to binocular, spectrally broad stimuli and analyze the spatial tuning of the PCoR, we performed experiments in a custom-built immersive dome illuminated by a digital light processing (DLP) projector via a spherical mirror (Figure 1F). The DLP projector provided white light stimulation covering the full complement of mouse photoreceptors, including the short-wavelength-sensitive cones (S-cones).[59] Projections spanned 120° in elevation and 180° in azimuth. Mice were head-fixed and free to run on a cylindrical treadmill inside the dome. Because the pupil diameter of head-fixed mice is correlated with their running speed,[60,61] we analyzed pupil responses to visual stimuli when the mice were standing still. The intensity dependence of the PLR (Figure 1G) and temporal-frequency tuning of the PCoR in the immersive dome (Figure 1H) matched those measured with LED stimulation. We next presented sign-inverting gratings (5 Hz) of varying spatial frequencies (0.001–0.1 cycles per degree or cpd). The PCoR exhibited low-pass spatial tuning, with robust responses restricted to spatial frequencies $\leq$ 0.01 cpd (Figure 1J). Comparisons of pupil diameters during running in the presence vs. absence of contrast stimuli showed that the PCoR can drive pupil constriction during high-arousal states (Figure 1I).

Rod phototransduction drives the PCoR

There are three classes of photoreceptors in the mammalian retina: rods, cones, and ipRGCs (Figure 2A). To understand which of them drives the PCoR, we selectively eliminated their phototransduction. Removal of GNAT1 (i.e., Gnat1 knockout [KO] mice),[62] the rod-specific α-transducin, abolished the PCoR (Figures 2B and 2C). Conversely, neither a hypomorphic allele of the cone-specific α-transducin GNAT2 (i.e., $Gnat2^{cpfl3}$ mice, see STAR Methods)[63] nor deletion of Opn4 (i.e., Opn4 KO mice),[19] the gene-encoding melanopsin, affected the PCoR (Figures 2B and 2C). Similarly, the intensity-response curves of the PLR were right-shifted in Gnat1 KO, but not in $Gnat2^{cpfl3}$ or Opn4 KO, mice (Figure 2D). In the PLR, melanopsin can support pupil constriction at high light levels.[41] Thus, the maximal pupil constriction for Gnat1 KO mice in the PLR at 10^5 R* did not differ from control mice at $10^{3.5}$ R*. By contrast, the PCoR depends strictly on rod phototransduction, and increasing the light intensity did not rescue the response in Gnat1 KO mice (Figure 2E).

A cell-type-specific retinal pathway mediates the PCoR

Rod signals, which drive the PCoR, travel from the outer to the inner retina through bipolar cells. There are 15 bipolar cell types in mice[34–36]; most are conserved in primates.[25,64,65] Anatomical studies showed that type 6 bipolar cell axons form en passant synapses with M1 ipRGC dendrites, suggesting that they may provide synaptic excitation for the PCoR.[31–33] However, the function of type 6 bipolar synapses with M1 ipRGCs has not been tested. We generated mice in which type 6 bipolar cells express channelrhodopsin-2 (i.e., $B6_{CCK}$-ChR2 mice) and labeled M1 ipRGCs in $B6_{CCK}$-ChR2 mice by retrograde tracer injections into the suprachiasmatic nucleus (Figure 3A) or crossing to a line in which ipRGCs express tdTomato (i.e., Opn4-tdT mice). We then targeted M1 ipRGCs for whole-cell patch-clamp recordings under two-photon guidance. We confirmed our targeting by including a spectrally separable fluorescent dye in the intracellular solution and analyzing dendritic morphologies in two-photon image stacks acquired at the end of each recording. When we blocked synaptic transmission from rods and cones to bipolar cells, optogenetic stimulation elicited fast excitatory postsynaptic currents (EPSCs) in M1 ipRGCs of $B6_{CCK}$-ChR2 mice but not in control mice without ChR2 (Figure 3B). The slower EPSCs elicited in control M1 ipRGCs without blockers reflect the cumulative delays of rod phototransduction and synaptic transmission compared with the direct bipolar cell activation in $B6_{CCK}$-ChR2 mice (Figure 3B).

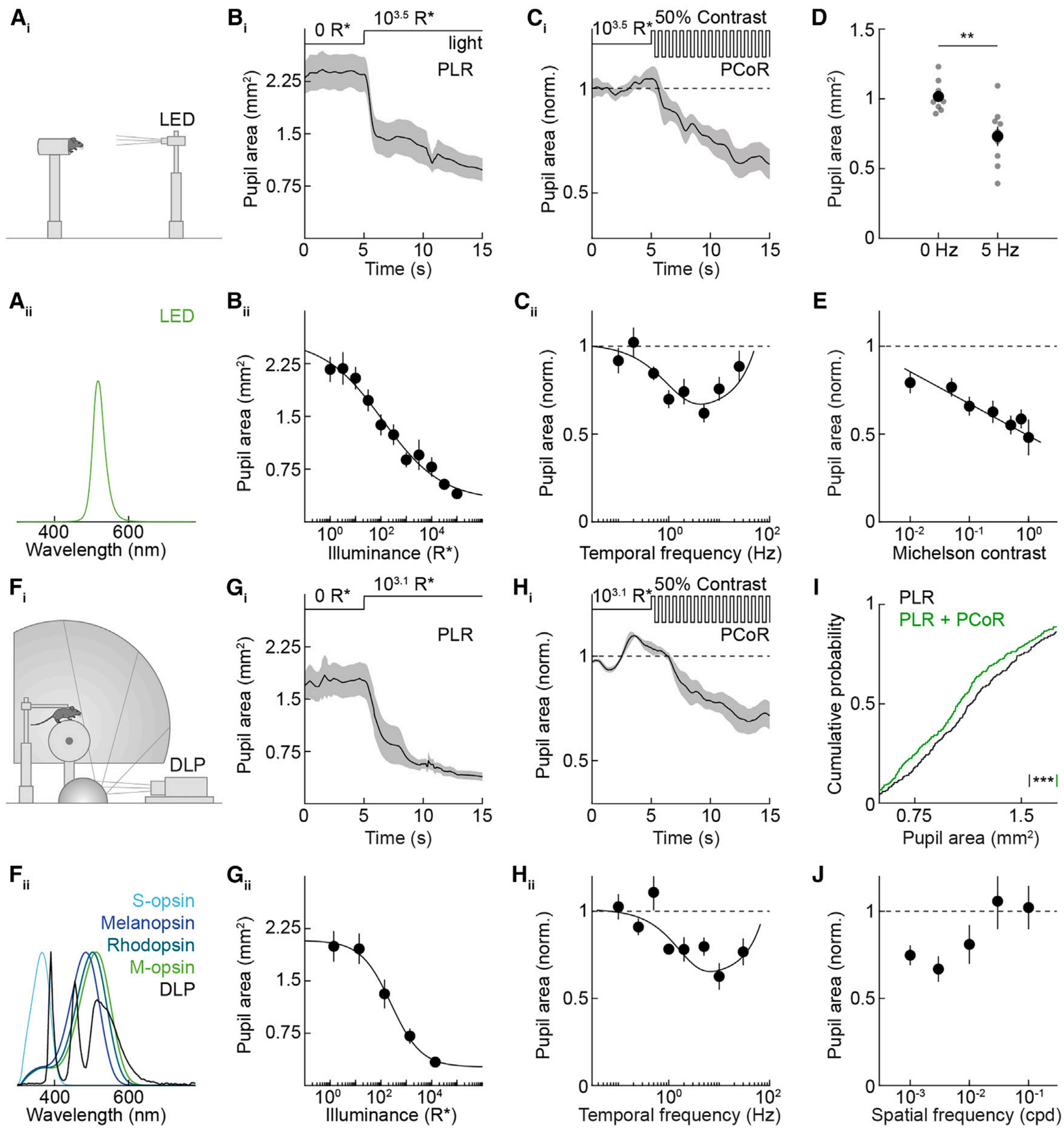


Figure 1. Pupil responses to luminance and contrast

(A) Schematic (i) of the behavioral apparatus used to assess pupillary light responses and the spectral distribution (ii) of the LED stimulus.

(B) Average trace (±SEM) (i) at $10^{3.5}$ R* and summary data (ii) of the pupillary light response (PLR) (n = 9).

(C) Average trace (± SEM) (i) at $10^{3.5}$ R*, 50% contrast, 5 Hz, and summary data (ii) of the pupillary contrast response (PCoR) (n = 22).

(D) Pupil area (in mm^2) under steady illuminance to a 50%-contrast light step (0 Hz) compared with 50% contrast modulation (5 Hz) (0 Hz: 1.02 ± 0.04 mm^2, n = 8; 5 Hz 0.73 ± 0.07 mm^2, n = 9; p = 0.004 by two-sample t test).

(E) Contrast dependence of the PCoR at 5 Hz (n = 9).

(F) Schematic (i) of the behavioral apparatus used to assess pupillary light responses under global illuminance conditions and the spectral distribution (ii) of the DLP stimulus (black) overlaid with the scaled spectra of all mouse photoreceptors (S-opsin, M-opsin, melanopsin, and rhodopsin).

(G) Average trace (±SEM) (i) at $10^{3.1}$ R* and summary data (ii) of the PLR (n = 8).

(H) Average trace (±SEM) (i) at $10^{3.1}$ R*, 50% contrast, 5 Hz global stimulation, and summary data (ii) of the PCoR (n = 13).

(I) Cumulative probability distribution of pupil areas during locomotion (>0.5 cm/s) either under the PCoR stimulus (green) or under steady illumination (black) (n = 13; p < 0.001 by Kolmogorov-Smirnov two-sample test).

(J) Spatial frequency dependence of the PCoR at 5 Hz (n = 13).

See also Figures S1 and S2.

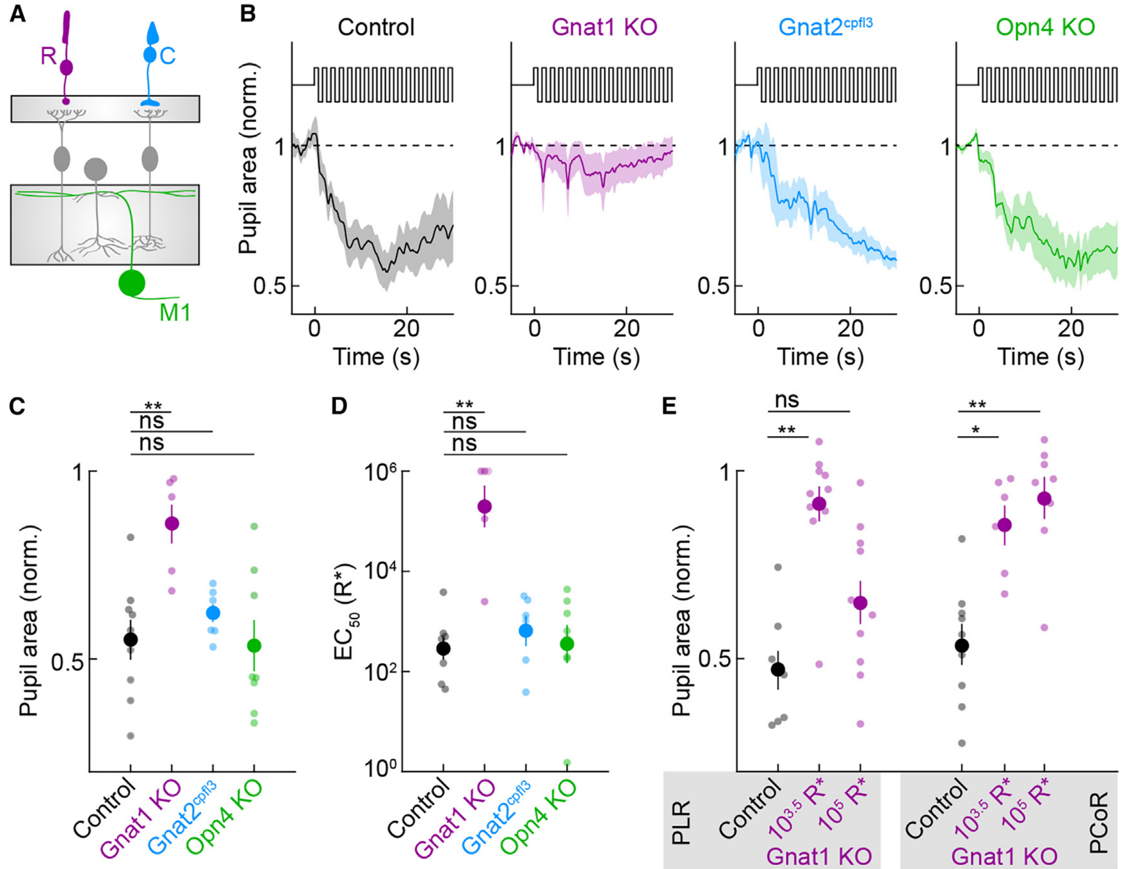


Figure 2. Photoreceptor elements of the PCoR

(A) Schematic diagram of the retina highlighting manipulations to rods (magenta, Gnat1 KO), cones (cyan, Gnat2^{cpfl3}), or melanopsin (green, Opn4 KO).

(B) Average PCoR traces (±SEM) at $10^{3.5}$ R*, 50% contrast, 5 Hz for control (n = 9), Gnat1 KO (n = 6), Gnat2^{cpfl3} (n = 7), and Opn4 KO (n = 8).

(C) PCoR relative pupil constriction at $10^{3.5}$ R*, 50% contrast, 5 Hz for control (0.55 ± 0.05, n = 9), Gnat1 KO (0.8587 ± 0.0514, n = 6), Gnat2^{cpfl3} (0.62 ± 0.02, n = 7), Opn4 KO (0.54 ± 0.07, n = 8) (Kruskal-Wallis effect of genotype p = 0.006; control vs. Gnat1 KO p = 0.0073, control vs. Gnat2^{cpfl3} p = 0.77, control vs. Opn4 KO p = 0.99).

(D) EC_{50} measured from the PLR for control ($10^{2.46 \pm 0.22}$ R*, n = 8), Gnat1 KO ($10^{5.30 \pm 0.42}$ R*, n = 6), Gnat2^{cpfl3} ($10^{2.8 \pm 0.31}$ R*, n = 6), Opn4 KO ($10^{2.55 \pm 0.38}$ R*, n = 8) (Kruskal-Wallis effect of genotype p = 0.006; control vs. Gnat1 KO p = 0.0044, control vs. Gnat2^{cpfl3} p = 0.85, control vs. Opn4 KO p = 0.91).

(E) Relative pupil area for PLR (left) and PCoR (right) responses for control at $10^{3.5}$ R*, Gnat1 KO at $10^{3.5}$ R*, and Gnat1 KO at 10^5 R* (PLR: Kruskal-Wallis effect of group p = 0.0003; control vs. Gnat1 KO at $10^{3.5}$ R* p = 0.0003, control vs. Gnat1 KO at 10^5 R* p = 0.2869, Gnat1 KO at $10^{3.5}$ R* vs. Gnat1 KO at 10^5 R* p = 0.242) (PCoR: Kruskal-Wallis effect of group p = 0.0013; control vs. Gnat1 KO at $10^{3.5}$ R* p = 0.0279, control vs. Gnat1 KO at 10^5 R* p = 0.0017, Gnat1 KO at $10^{3.5}$ R* vs. Gnat1 KO at 10^5 R* p = 0.82.

The amplitudes of the optogenetically evoked type 6 bipolar inputs (Figure 3B) were close to the total photoreceptor-driven input of M1 ipRGCs,[66] indicating that type 6 bipolar cells dominate their synaptic excitation.

To test whether type-6-mediated synaptic excitation drives the PCoR (Figure 3C), we generated mice in which type 6 bipolar cells express the diphtheria toxin receptor (DTR) (i.e., $B6_{CCK}$-DTR mice, see STAR Methods). Binocular diphtheria toxin (DT) injections in adult $B6_{CCK}$-DTR mice selectively removed type 6 bipolar cells, leaving the rest of the retina intact (Figures S3B, S3C, S4E, and S4F, see STAR Methods), and abolished the PCoR (Figures 3D and 3E). The intensity-response functions of the PLR were right-shifted in $B6_{CCK}$-DTR mice (Figure 3F). As in Gnat1 KO mice, higher-intensity light steps in $B6_{CCK}$-DTR mice could drive similar maximal pupil constriction (PLR) as lower-intensity steps in control mice (Figure 3G). We observed no compensation in the PCoR (Figure 3G). The matching results from $B6_{CCK}$-DTR and Gnat1 KO mice suggest that type 6 bipolar cells convey the rod signals that drive the PCoR and highlight the differences between PCoR and PLR, which rely solely on synaptic excitation and combine synaptic and intrinsic excitation, respectively.

The dendrites of type 6 bipolar cells exclusively contact cones.[67,68] Therefore, rod signals could reach type 6 bipolar cells either through the primary rod pathway in which rods connect to rod bipolar cells, which excite AII amacrine cells, which in turn form electrical synapses with type 6 axons, or the secondary rod pathway in which rods are electrically coupled to cones (Figure 3C).[69,70] We generated mice in which cones express the DTR (i.e., cone-DTR mice, see STAR Methods) and ablated cones in adult mice with intraperitoneal DT injections (Figures S4B and

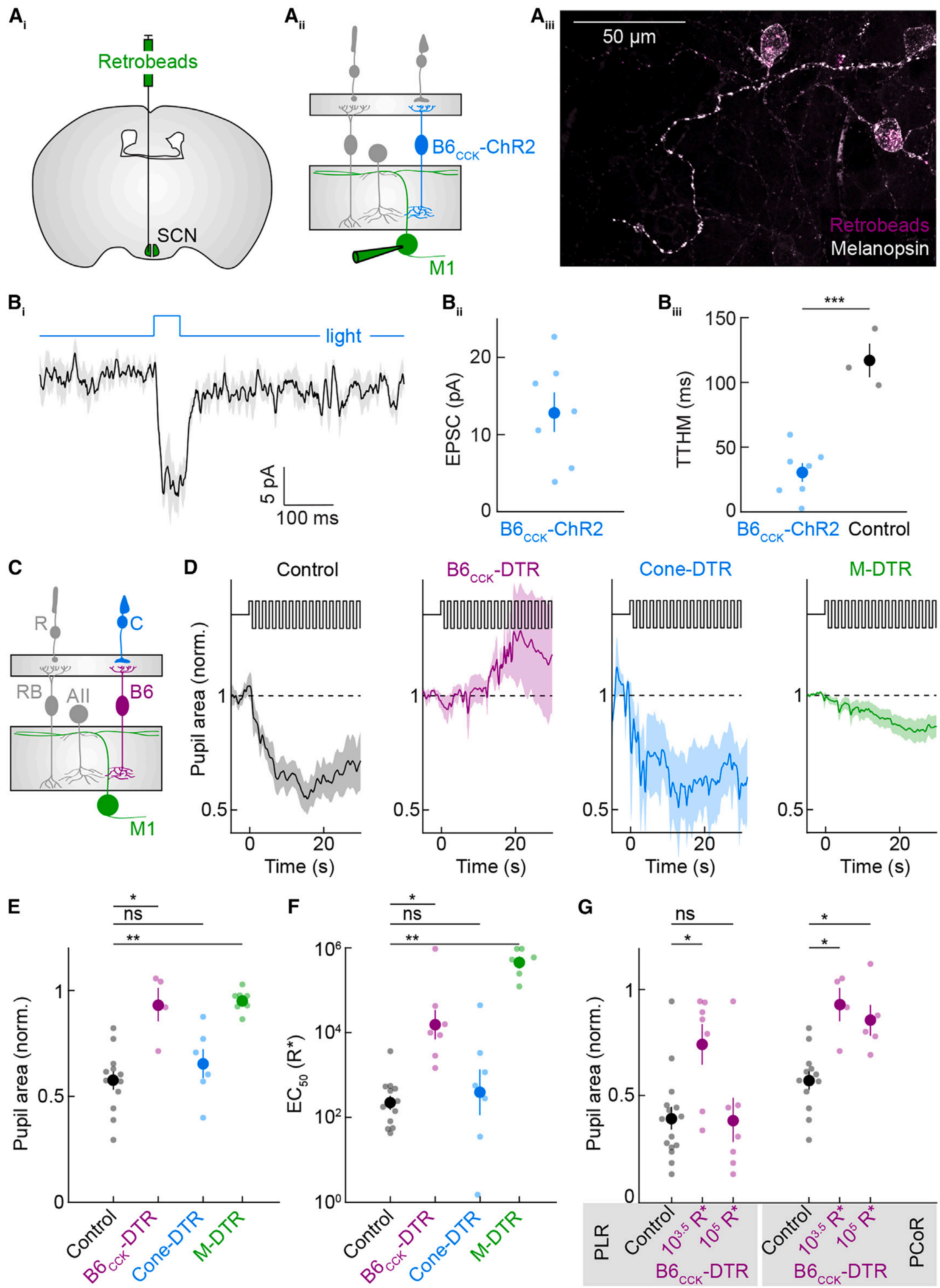


Figure 3. Circuit elements of the PCoR

(A) Strategy for labeling M1 ipRGCs and recording the optogenetically evoked input from type 6 bipolar cells (B6). Fluorescent retrobeads were injected bilaterally into the SCN (i), and M1 ipRGCs were targeted in the $B6_{CCK}$-ChR2 mouse line, which expresses channelrhodopsin-2 in all type 6 bipolar cells (ii). Fluorescent retrobead-positive cells were also positive for melanopsin (iii, scale bars, 50 μm).

(legend continued on next page)

S4C, see STAR Methods).[71,72] Cone removal neither affected the PCoR nor the PLR (Figures 3D–3F), indicating that the secondary pathway is dispensable and that the rod signals driving the PCoR likely reach type 6 bipolar cells through the primary rod pathway.

The PLR is mediated by M1 ipRGCs projecting to the OPN shell.[16,18,73,74] We generated mice in which ipRGCs express the DTR (i.e., M-DTR mice, see STAR Methods) and removed these cells in adult animals (Figures S3B, S3C, and S4G–S4K, see STAR Methods). In addition to the PLR, this nearly completely abolished the PCoR, suggesting that ipRGCs (specifically the M1 type) are the conduit for both responses (Figures 3D–3F).[16,18] In addition to shell-projecting M1 ipRGCs, DT injections in M-DTR mice also ablated M2 ipRGCs and removed a fraction of M4 ipRGCs (Figures S4I–S4N), both of which innervate the OPN core.[73] The role of the OPN core in pupillary responses is unknown.[75,76] Retinal projections to the OPN core were grossly intact in DT-injected M-DTR mice, including the M6 ipRGCs (Figure S5), which dominate input to the OPN core.[76] Together, these experiments delineate a cell-type-specific pathway from rods to type 6 bipolar cells (likely via rod bipolar and AII amacrine cells) and M1 ipRGCs that mediate the PCoR.

The site and mechanism of the PCoR computation

To drive pupil constriction to contrast, neurons in the PCoR pathway must convert stimulus fluctuations into sustained responses. To localize the site and understand the mechanisms of this conversion, we used two-photon imaging and patch-clamp recordings to measure responses at different stages of the PCoR pathway (Figure 4). We presented full-field stimuli square-wave modulated (50% contrast) at different temporal frequencies. We analyzed responses at the stimulus frequencies (F1) and sustained responses (F0) to see where and how the conversion arises.

We started our investigation by measuring calcium dynamics in type 6 bipolar cell axons. We intraocularly injected adeno-associated viruses expressing Cre in ON bipolar cells (AAV-Grm6-Cre) into Cre-dependent GCaMP6f reporter mice (Ai148).[77,78] This strategy produced sparse and intense GCaMP6f expression in bipolar cells. We identified type 6 bipolar cells by their axonal stratification and dendritic connectivity patterns.[36,67,68] The calcium signals in type 6 bipolar cell axons tracked stimulus fluctuations across frequencies (Figure 4A) without sustained responses (Figure 4A), indicating that the site of conversion lies downstream.

We next targeted M1 ipRGCs, labeled genetically or by retrograde tracing, for cell-attached and whole-cell patch-clamp recordings. M1 ipRGC EPSCs elicited by temporal contrast showed robust F1 but no F0 responses to stimulus frequencies driving the PCoR (Figure 4B). Thus, the conversion of fluctuating stimuli into sustained responses does not occur in the synaptic transmission of signals from type 6 bipolar cells to M1 ipRGCs.

However, M1 ipRGC spike trains elicited by temporal contrast stimuli exhibited both F1 and F0 responses (Figures 4C and 4D). The frequency tuning of the F0 responses matched that of the PCoR (Figure 4C), independent of the recording configuration (cell attached vs. whole cell) or method of targeting M1 ipRGCs (genetically labeled vs. retrogradely traced) (Figures S6A–S6C). The frequency tuning of the M1 ipRGC's F0 response is not a universal feature of ganglion cell responses to temporal contrast stimuli; M4 ipRGCs (i.e., sONα RGCs) exhibit a starkly different tuning curve (Figure S6D). Furthermore, like the PCoR, the increase in the sustained firing of M1 ipRGCs to a 50%-contrast flicker (5 Hz) exceeded that elicited by a 50%-contrast step (normalized firing rate 50%-contrast step: 0.05 ± 0.07, $n = 4$; 50%-contrast flicker: 0.78 ± 0.08, $n = 13$; $p < 0.001$ by two-sample t test), even though the luminance of the step stimulus is 1.5-fold greater. When we compared the relationship of the pupil area with M1 ipRGCs' F0 responses for the PLR and the PCoR, we found that this relationship was approximately linear and that the data points for the PLR and PCoR fell within the same envelope (Figure 4E). Finally, we found that M1 ipRGC spike trains elicited by square-wave current injections at different temporal frequencies had strikingly similar F0 and F1 tuning profiles to the spike trains elicited by visual stimuli (Figure 4F). Together, these results indicate that the conversion of temporal contrast into sustained responses occurs in the M1 ipRGCs' translation of synaptic excitation into spike responses, with no evidence for differential processing of luminance and contrast signals downstream.

The pupil's effects on retinal image brightness and contrast in mice

To explore why the pupil may respond to both luminance (PLR) and contrast (PCoR), we quantified the effects of pupil constriction on the retinal brightness and contrast in mice.

(B) Average trace (±SEM) (i) for M1 ipRGC EPSCs in response to optogenetic stimuli and summary data (ii, 12.90 ± 2.56 pA, $n = 7$). Comparison of time to half maximum (TTHM) response (in ms) for optogenetic responses in the presence of synaptic blockers vs. photoreceptor responses in control animals without synaptic blockers ($p < 0.001$ by two-sample t test).

(C) Schematic diagram of the retina highlighting manipulations to type 6 bipolar cells (magenta, $B6_{CCK}$-DTR), cones, and the secondary rod pathway (cyan, cone-DTR), or M1 ipRGCs (green, M-DTR).

(D) Average PCoR traces (±SEM) at $10^{3.5}$ R*, 50% contrast, 5 Hz for control ($n = 9$), $B6_{CCK}$-DTR ($n = 4$), cone-DTR ($n = 6$), and M-DTR ($n = 8$).

(E) PCoR relative pupil constriction at $10^{3.5}$ R*, 50% contrast, 5 Hz for control (0.55 ± 0.05, $n = 9$), $B6_{CCK}$-DTR (0.94 ± 0.08, $n = 4$), cone-DTR (0.66 ± 0.07, $n = 6$), M-DTR (0.95 ± 0.02, $n = 8$) (Kruskal-Wallis effect of genotype $p = 0.007$; control vs. $B6_{CCK}$-DTR $p = 0.0182$, control vs. cone-DTR $p = 0.9125$, control vs. M-DTR $p = 0.0029$).

(F) EC_{50} measured from the PLR for control (2.46 ± 0.22, $n = 8$), $B6_{CCK}$-DTR (4.22 ± 0.35, $n = 7$), cone-DTR (2.61 ± 0.54, $n = 7$), M-DTR (5.68 ± 0.14, $n = 8$) (Kruskal-Wallis effect of genotype $p = 0.0001$; control vs. $B6_{CCK}$-DTR $p = 0.0193$, control vs. cone-DTR $p = 0.8867$, control vs. M-DTR $p = 0.0002$).

(G) Relative pupil area for PLR (left) and PCoR (right) responses for control at $10^{3.5}$ R*, $B6_{CCK}$-DTR at $10^{3.5}$ R*, and $B6_{CCK}$-DTR at 10^5 R* (PLR: Kruskal-Wallis effect of group $p = 0.043$; control vs. $B6_{CCK}$-DTR at $10^{3.5}$ R* $p = 0.05$, control vs. $B6_{CCK}$-DTR at 10^5 R* $p = 0.9759$, $B6_{CCK}$-DTR at $10^{3.5}$ R* vs. $B6_{CCK}$-DTR at 10^5 R* $p = 0.0797$) (PCoR: Kruskal-Wallis effect of group $p = 0.0027$; control vs. $B6_{CCK}$-DTR at $10^{3.5}$ R* $p = 0.0118$, control vs. $B6_{CCK}$-DTR at 10^5 R* $p = 0.0250$, $B6_{CCK}$-DTR at $10^{3.5}$ R* vs. $B6_{CCK}$-DTR at 10^5 R* $p = 0.9170$).

See also Figures S3–S5.

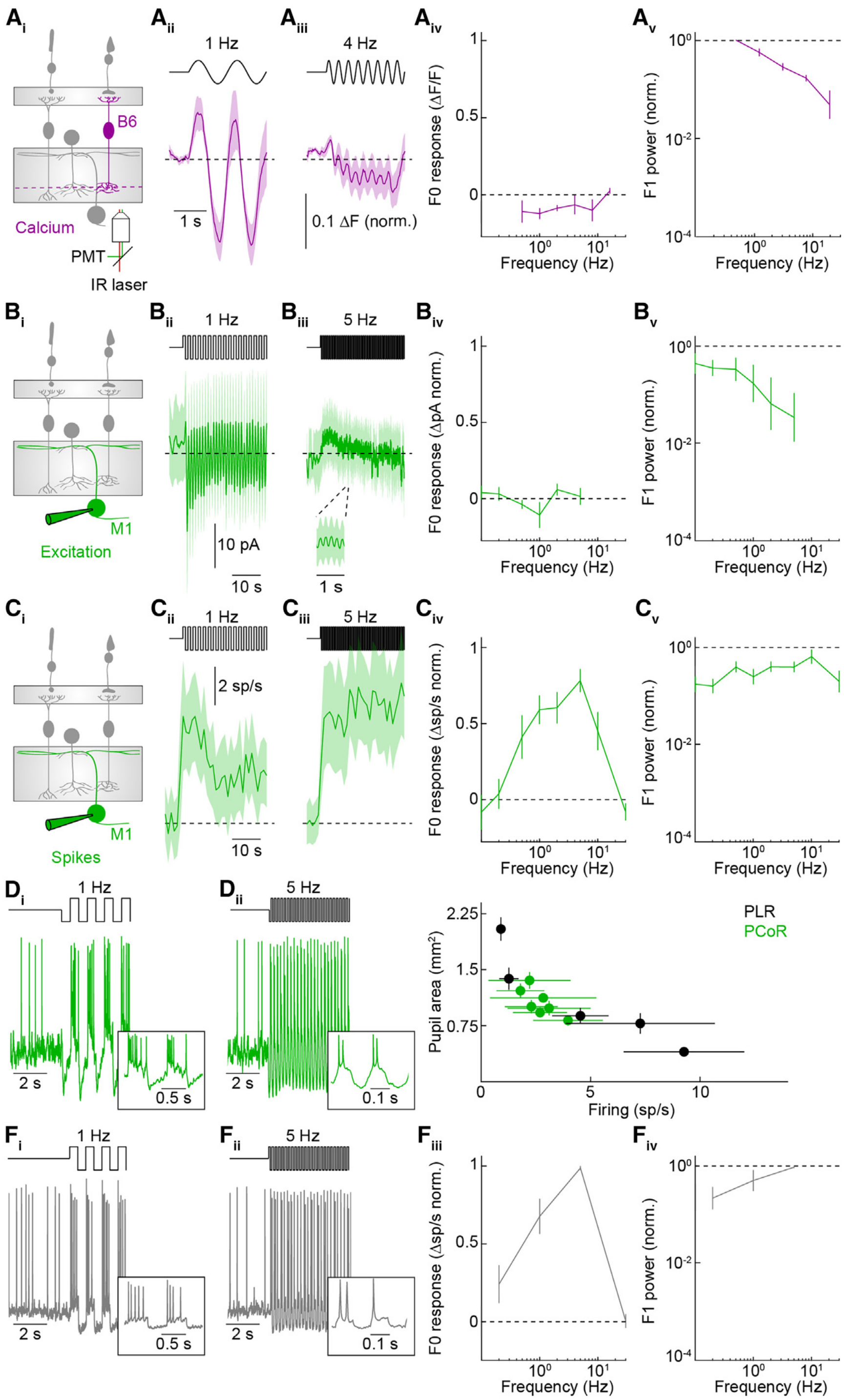


Figure 4. Mechanism of the PCoR computation

(A) Calcium imaging of type 6 bipolar cell axon terminals. Schematic of the retina with targeted cell type and recording strategy (i). Average traces (±SEM) at low (ii) and high (iii) temporal frequencies. F0 (iv) and F1 (v) responses at each tested frequency ($n = 6$).

(B) Analogous to (A), but for EPSCs from M1 ipRGCs ($n = 4$).

(C) Analogous to (A), but for spike rates recorded from M1 ipRGCs ($n = 13$).

(D) Example spike traces at low (ii) and high (iii) temporal frequencies, with insets to show two response cycles.

(E) Correlation of M1 ipRGC firing rate obtained during electrophysiology recordings ($n = 6$), with the pupil area observed during behavioral experiments ($n = 9$) for illuminance steps from darkness (1, 2, 3, 4, and 5 $\log_{10}$ R*, black) and light-adapted (3.5 $\log_{10}$ R*) contrast modulation (0.1, 0.2, 0.5, 1, 2, 5, and 10 Hz, green).

(F) Example spike traces to square-wave current injection at low (i) and high (ii) temporal frequencies, with insets to show two response cycles. F0 (iii) and F1 (iv) responses at each tested frequency ($n = 4$).

See also Figure S6.

Retinal illuminance varies proportionally to the square of the pupil radius, such that halving the pupil diameter cuts the light admitted into the eye to one-quarter of its intensity (Figure 5A). Considering that ambient light levels vary by a factor of $\sim10^9$ from moonless nights to sunny days[79] and that the PLR is distributed over a range of 10^4 brightness units, the reduction in retinal light intensity provided by maximal pupil constriction (a factor of $\sim10^{0.5}$) is unlikely to make major contributions to light adaptation. Furthermore, light passing through the pupil center (including during pupil constriction) more effectively activates cone photoreceptors and is perceived more brightly than light passing near the edge of a dilated pupil. This phenomenon, known as the Stiles-Crawford effect (SCE),[80–82] results largely from mitochondrial microlenses in cones rejecting off-angle light.[83] Including the SCE in our calculations further reduced the impact of pupil constriction on the effective retinal image brightness (Figure 5B).

The dependence of spatial contrast on the pupil radius is more complicated than that of illuminance. Any optical system, including the eye, can be described by an optical transfer function that specifies how well the system transmits each spatial frequency up to the diffraction limit. This transfer is approximately equivalent to the perceived contrast of a 100%-contrast object at a given spatial frequency. Using the Image System Engineering Toolbox for Biology (ISETBIO),[84–86] we modeled the effect of pupil radius on the optical transfer function of the mouse eye at wavelengths reflecting the spectral filtering of rods (Figure 5C) or M-cones (Figure 5D).[59,87,88] In both cases, the low spatial input frequencies (e.g., 0.1 cpd) are transmitted to the retinal image with maximum efficiency for all pupil diameters. However, near

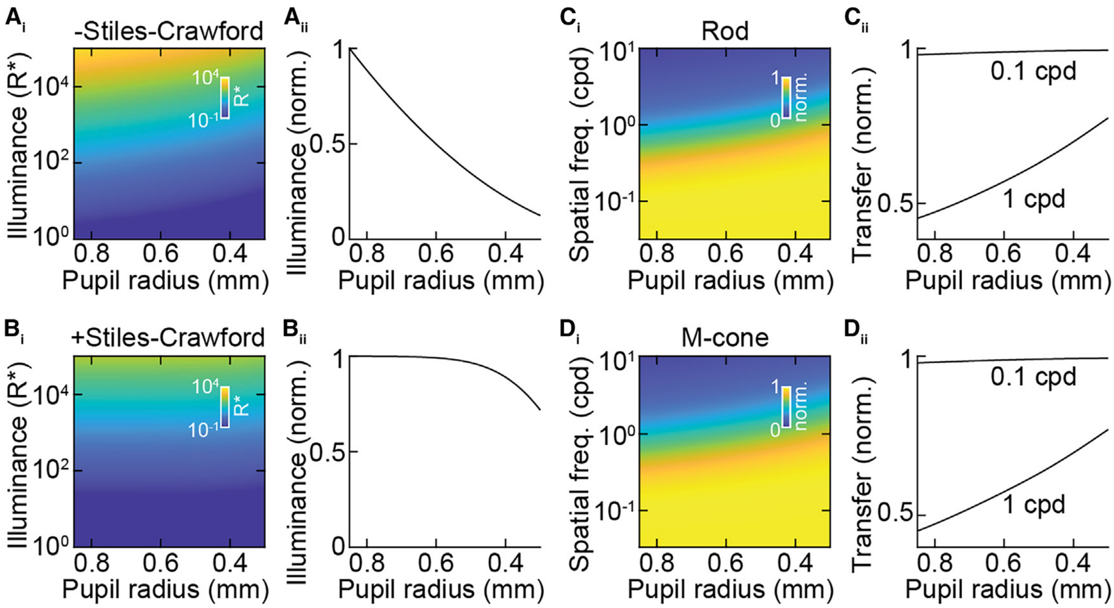


Figure 5. Pupil effect on illuminance and contrast

(A) Dependence of the absolute retinal illuminance (Stiles-Crawford effect [−SCE], color coded) on corneal illuminance and pupil size (i) and relative retinal illuminance on pupil size alone (ii).

(B) Dependence of the effective retinal illuminance (i.e., the retinal illuminance for cone vision considering the SCE, +SCE, color coded) on corneal illuminance and pupil size (i) and relative effective retinal illuminance on pupil size alone (ii).

(C) Relative optical transfer (color coded) as a function of spatial frequency and pupil size (i), with direct comparisons of 0.1 and 1 cpd (ii) for rod-weighted vision.

(D) Analogous to (C), but for M-cone-weighted vision.

the limit of mouse visual acuity (~1 cpd),[89] pupil constriction dramatically increases spatial contrast in the retinal image (from ~40% to ~80% transmission, Figures 5C and 5D). Thus, pupil constriction likely contributes only marginally to light adaptation but could greatly enhance contrast sensitivity and visual acuity.

Pupil constriction enhances contrast sensitivity and acuity in mice

To test whether the increased high-spatial-frequency contrast in retinal images transmitted through constricted pupils increases contrast sensitivity and acuity in mice, we analyzed two visual behaviors: the gaze-stabilizing optokinetic reflex and predation.

To gauge the impact of the pupil on the contrast sensitivity and acuity of the gaze-stabilizing optokinetic reflex, we tracked eye movements elicited by drifting gratings of varying contrasts and spatial frequencies in awake head-fixed mice (Figures 6A and 6B). All mice were light-adapted, and the drifting grating stimuli were luminance neutral, minimizing the contributions of pupillary illuminance control to our results. The optokinetic reflex is restricted to spatial frequencies whose contrast transmission is enhanced by pupil constriction (Figure 6D).[90] The optokinetic reflex was drastically reduced in both M-DTR mice, which have dilated pupils from a loss of ipRGCs, and control mice given atropine eye drops to pharmacologically induce pupil dilation without affecting the retina (Figures 6C–6E). Conversely, carbamylcholine (carbachol) eye drops induced maximal pupil constriction in M-DTR mice, reducing their pupil diameters beyond those of the untreated control mice (Figure 6C). This peripheral manipulation of the pupil restored the optokinetic reflex in M-DTR mice and elevated its contrast sensitivity and acuity (i.e., the highest spatial frequency grating eliciting eye tracking movements) above control mice (Figures 6D and 6E).

To analyze how the pupil influences complex visual behaviors, we focused on predation (Figure 6F), a major driver of visual system evolution that depends on acuity.[48,50,91–93] We tested cricket hunting in light-adapted control mice and mice with genetically (i.e., M-DTR) or pharmacologically-(i.e., atropine) induced pupil dilation. Using a published dataset,[6] we confirmed that mice do not change their pupil size with the ocular or locomotor movements accompanying hunting (Figure S7), suggesting that light-adapted pupil sizes (Figure 6C) remained relatively constant. In all conditions, mice kept crickets in their binocular visual fields while hunting (Figure 6G). Both M-DTR and atropine-treated control mice caught crickets more slowly than untreated control mice (Figure 6H). This reduction in performance was driven by deficits in prey detection, measured as the interval between subsequent approaches (Figure 6H), and pursuit, measured as the number of approaches needed to catch a cricket (Figure 6H). To avoid manipulating the eyes, we also injected aAAVs expressing a pharmacogenetic silencer (AAV-DREADDi) into the OPN, which mediates light-evoked pupil constriction (Figure 6I).[16,94,95] Intraperitoneal injections of clozapine-N-oxide (CNO), but not phosphate-buffered saline (PBS), dilated the pupils of these mice for a few hours (Figure 6J, pupil diameter at hunting light levels, PBS: 35.5% ± 4.0%, CNO: 65.0% ± 11.3%, p = 0.0175 by

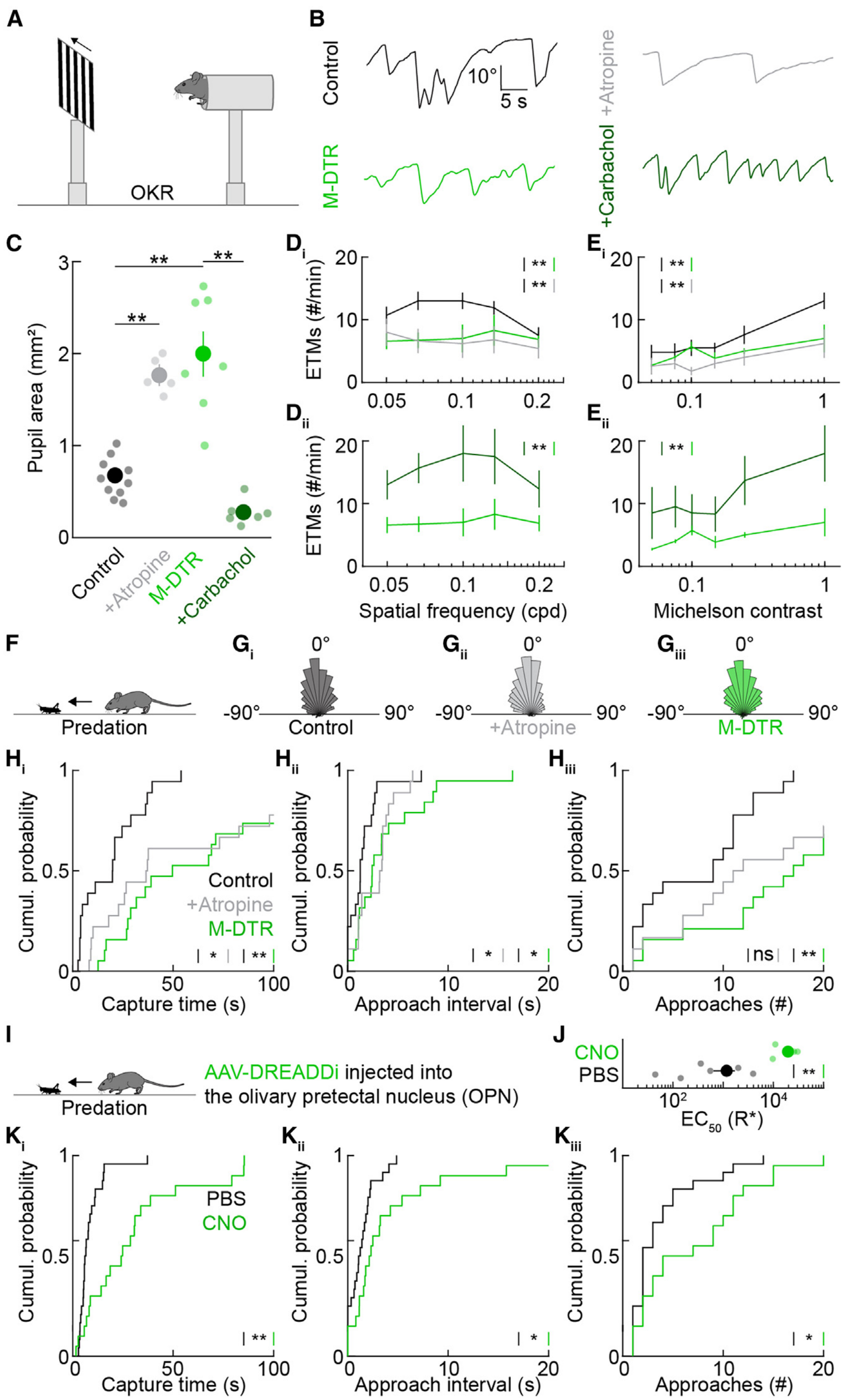


Figure 6. Pupil constriction enhances visual performance

(A) Schematic of the setup for recording optokinetic responses from head-fixed mice.

(B) Example optokinetic responses to drifting bars (0.1 cpd, Michelson contrast = 1) for control, control + atropine, M-DTR, or M-DTR + carbachol mice.

(C) Summary data of pupil size for control (n = 10), control + atropine (n = 5), M-DTR (n = 7), and M-DTR + carbachol (n = 6) mice (ANOVA effect of group $p < 0.0001$; control ± atropine $p = 0.0006$; control vs. M-DTR $p < 0.0001$; M-DTR ± carbachol $p < 0.0001$).

(D) Summary data of optokinetic responses in eye-tracking movements (ETMs) per minute across spatial frequencies (Michelson contrast = 1) comparing control, control + atropine, and M-DTR mice (i) or M-DTR mice ± carbachol (ii) (ANOVA effect of group $p < 0.0001$; control ± atropine $p = 0.0017$; control vs. M-DTR $p = 0.0018$; M-DTR ± carbachol $p < 0.0001$).

(E) Same as for (E), but for optokinetic responses across contrasts at 0.1 cpd (ANOVA effect of group $p < 0.0001$; control ± atropine $p = 0.0065$; control vs. M-DTR $p = 0.0061$; M-DTR ± carbachol $p < 0.0001$).

(F) Schematic of predation experiments in which mice hunt crickets.

(G) Histogram of azimuthal cricket location during approach periods for control (i, n = 5), control + atropine (ii, n = 5), and M-DTR (iii, n = 6) mice. (Bootstrap $p > 0.05$ between all conditions).

(H) Cricket hunting performance measured by the time to capture crickets (i, control: n = 20 crickets for 5 mice, control + atropine: n = 20 crickets for 5 mice, M-DTR: n = 21 crickets for 6 mice; control ± atropine $p = 0.0224$, control vs. M-DTR $p = 0.0011$ by Kruskal-Wallis test), the interval between approaches to the cricket (ii, control ± atropine $p = 0.0275$, control vs. M-DTR $p = 0.0296$), and the number of approaches made (iii, control ± atropine $p = 0.1131$, control vs. M-DTR $p = 0.0061$).

(I) Schematic of predation experiments in which mice injected with an AAV expressing a pharmacogenetic silencer into the olivary pretectal nucleus hunt crickets.

(J) Summary data of pupil sizes after PBS (n = 7) and CNO injections (n = 5, $p = 0.0025$).

(K) Cricket hunting performance measured by the time to capture crickets (i, PBS: n = 24 crickets for 8 mice, CNO: n = 20 crickets for 7 mice, $p = 0.001$, by Mann-Whitney U test), the interval between approaches to the cricket (ii, $p = 0.025$), and the number of approaches made (iii, $p = 0.045$).

See also Figure S7.

Mann-Whitney U test). CNO-injected mice caught crickets more slowly than PBS-injected mice and had deficits in prey detection and pursuit (Figure 6K).

Thus, the frequency-dependent contrast enhancement in the retinal image from pupil constriction increases the sensitivity and acuity of a gaze-stabilizing brainstem reflex and improves performance in a complex visual survival behavior.

The PCoR sharpens retinal images beyond the luminance-dependent constriction

Our modeling and behavioral results indicate that the pupil's control of image contrast may be more important than its influence on image brightness, as previously hypothesized.[12,13] To estimate the specific contribution of the PCoR to this regulation, we modeled the optical transfer function at different pupil sizes

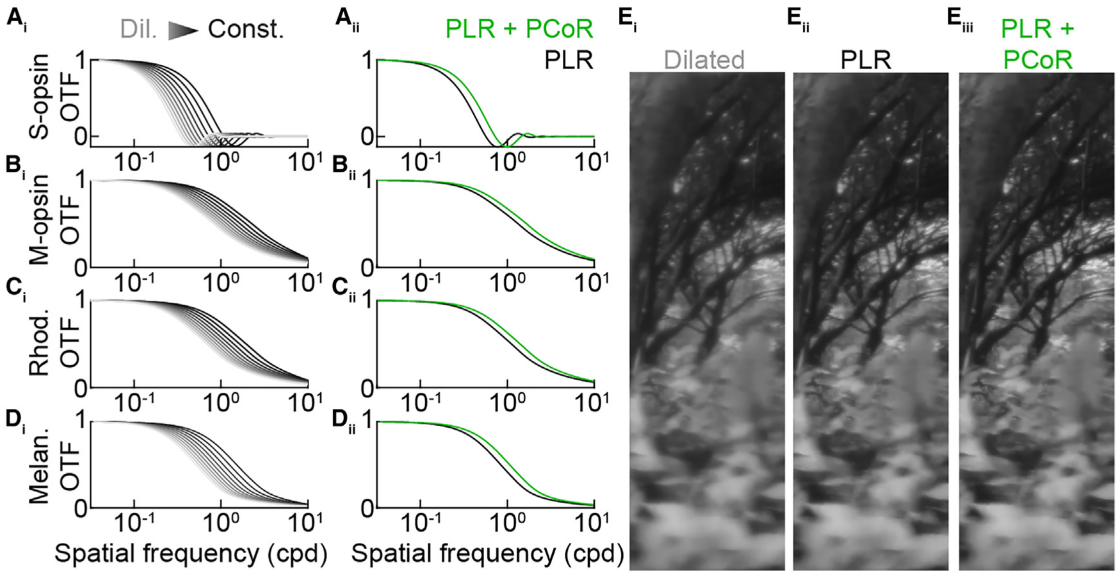


Figure 7. Effects of pupil size on the retinal image

(A) Optical transfer as a function of pupil size (i) across the physiological range of mouse pupil radii from dilated (gray) to constricted (black) weighted for S-opsin. Experimentally determined pupil radii for an illuminance step (PLR, black) vs. contrast modulation at that illuminance (PCoR, green) exhibit significantly different curves ($p < 0.0001$ by bootstrapping).

(B) Analogous to (A) for M-opsin.

(C) Analogous to (A) for rhodopsin.

(D) Analogous to (A) for melanopsin.

(E) Comparison of the retinal image for a dilated mouse pupil (i), illuminance constricted pupil (ii), and contrast constricted pupil (iii).

for wavelengths weighted for S-cones, M-cones, rods, and melanopsin (Figures 7A–7D). We then compared the optical transfer functions of the light-adapted pupil sizes (i.e., PLR) with those induced by temporal contrast at the same mean luminance (i.e., PLR + PCoR) measured in our behavioral experiments (Figure 1). The PCoR significantly improved the optical transfer at higher spatial frequencies over the PLR alone (Figures 7A–7D), resulting in sharper retinal images compared with the dilated pupils of pupils constricted by luminance alone (Figure 7E).

The PCoR in humans

Mice and humans explore unique visual environments with different viewing strategies. We recruited seven volunteers to test whether the PCoR is present in humans and to analyze its tuning properties. We presented the same temporal contrast stimuli we had shown mice on a computer monitor while recording the volunteers' pupils with an infrared camera (Figure 8A). We also tested the classic PLR response (Figure 8B). Thus, we found that humans have a robust PCoR (Figures 8C and S8). The human PCoR exhibited bandpass temporal tuning with strong responses from 0.2 to 5 Hz (Figure 8C).

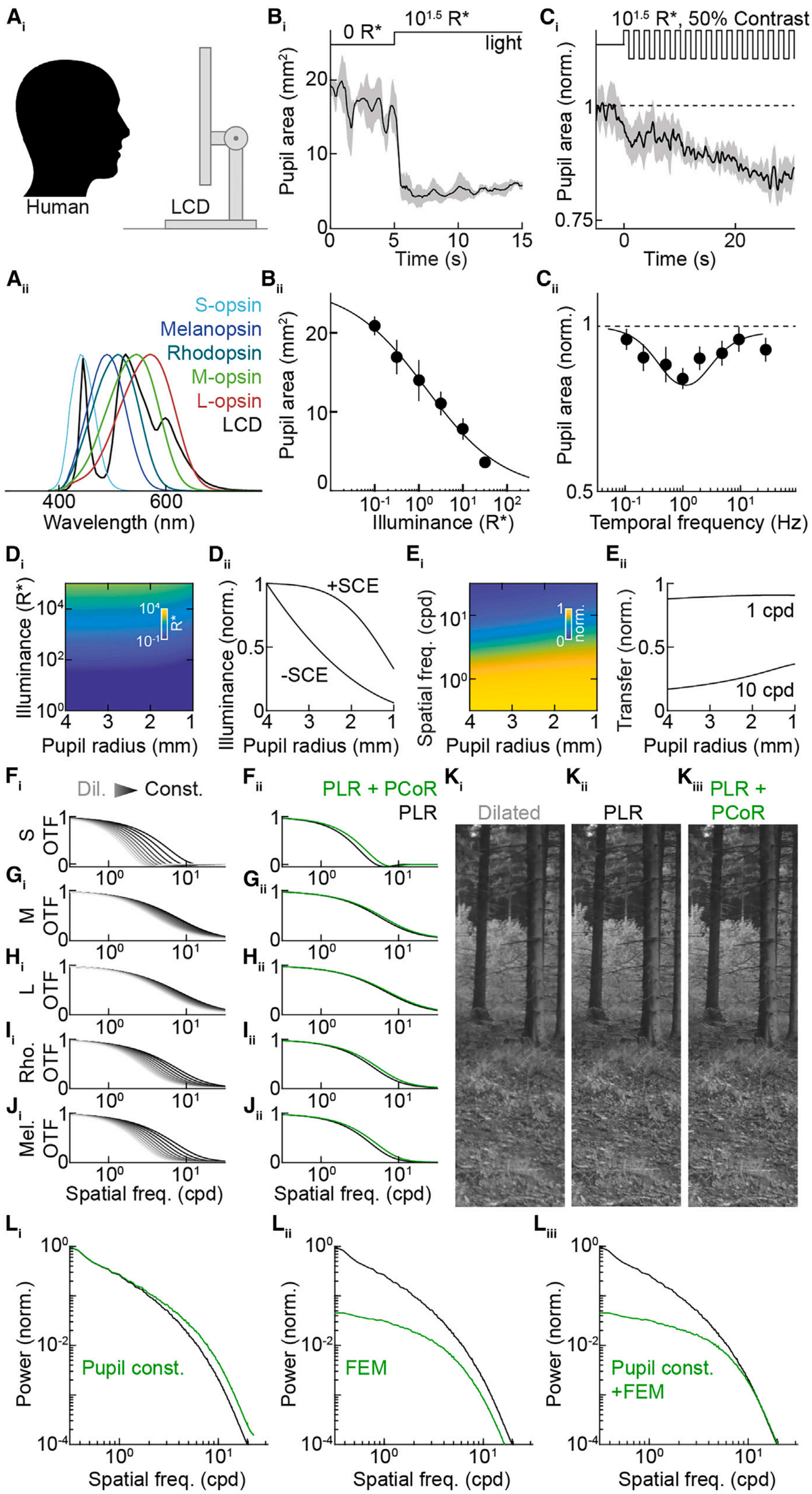
As in mice, computational modeling suggested that pupil constriction in humans has a limited effect on retinal illuminance (Figure 8D). For example, a constriction from 4 to 1 mm in pupil radius reduces retinal illuminance by a factor of $10^{1.15}$ compared with the range of $\sim 10^9$ in environmental brightness[79] (Figure 8D). Considering the SCE, the effective retinal illuminance and perceived brightness are attenuated even less (by a factor of $10^{0.4}$, Figure 8D). We next modeled the effect of pupil constriction on the optical transfer function matching the spectral filtering of M-cones (Figure 8E). Low spatial input frequencies were transmitted independent of pupil size, but high spatial frequencies were transmitted optimally at smaller pupil sizes (Figures 8F–8J). Compared with mice, the impact of the pupil on the retinal image was shifted to higher spatial frequencies, matching the higher acuity of humans (Figures 8F–8J). Finally, the constriction due to the contrast response (i.e., PCoR + PLR) provided additional improvements over the steady-state illuminance pupil constriction alone (PLR) for wavelengths weighted for S-cones, M-cones, L-cones, rods, and melanopsin, leading to sharper retinal images (Figure 8K).

DISCUSSION

Complex pupil responses in mice and humans

We discovered that the pupil responds robustly to contrast (PCoR) in mice (Figure 1) and humans (Figure 8). The PCoR exhibits temporal band-pass (mouse: 0.5–10 Hz, humans: 0.2–5 Hz) and spatial long-pass (mouse: ≤ 0.01 cpd) tuning and scales linearly with the logarithm of contrast (Figure 1). The PCoR's spatiotemporal tuning matches the size and frequency of gaze-shifting head and eye movements (i.e., saccades) in mice and humans.[4–7,96] Therefore, we hypothesize that the PCoR adjusts the pupil to contrast in the environment sampled by saccades.

Pupil responses to temporal flicker have been observed in humans and non-human primates.[97–101] However, these studies employed stimuli that were not luminance neutral and focused



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their analyses on the pupillary F1 responses observed at lower temporal flicker frequencies.[97–101] Instead, the PCoR refers to the sustained pupil constriction (i.e., the F0 response) elicited by luminance-neutral contrast stimuli.

In addition to contrast, the pupil responds to luminance (PLR). This dual input sensitivity matches the dual output functions of the pupil, which control retinal illuminance and contrast (Figure 5). Luminance and contrast vary independently in natural visual scenes.[102,103] Together with the output duality, this statistical independence may explain why the pupillary reflex arc measures luminance and contrast separately.

The pupil also constricts as part of a motor program, including lens accommodation and vergence eye movements, activated when subjects focus on near objects (i.e., the pupillary near response or PNR), increasing the depth of focus.[11,104] Conversely, the pupil dilates during psychological processes, including arousal and motor preparation (i.e., the psychosensory pupil response or PPR).[105] Our data indicate that PCoR can drive pupil constriction across arousal states in mice (Figure 1), and human PPRs tend to be smaller (Δ pupil diameter: $\sim$0.1 mm)[105–107] than the PCoR we measured in volunteers (Δ pupil diameter: 0.75 ± 0.25 mm, Figure 8). How the PCoR, PLR, PNR, and PPR, all of which are adaptive in isolation, cooperate to optimize vision in complex environments is an intriguing question for future studies.

Pupil control of the retinal image and visual behavior

The pupil regulates both retinal illuminance and image contrast; the former is more widely appreciated, but the latter, we argue, is more significant. We combine computational modeling, pathway-specific manipulations, and behavioral observations to support this conclusion. Computational modeling considering the optical aberrations of mouse and human eyes[84–86] indicated that, in both species, the pupil's impact on retinal illuminance and image contrast is comparable (Figures 5 and 8). Thus, relative to their input values, the pupil can adjust brightness and spatial contrast by a factor of $\sim 10^{0.5}$. However, whereas the retina functions over a vast range of illuminances (operating range: $\sim 10^9$), contrast encoding in the retina and downstream is more limited (operating range: $<10^2$).[79,108–110] Thus, compared with the operating ranges of the visual system, the pupil's impact on contrast is far greater ($>10^7$-fold) than its impact on brightness, which is further reduced by the SCE.[80–83] Nonetheless, the limited pupil-mediated changes in illuminance can shape the kinetics of dark adaptation and poise the retina for specific functions through luminance-dependent switches in its circuits.[35,111]

The pupil's effect on retinal image contrast depends on the spatial frequency. Whereas low-frequency contrast is transmitted from the input to the retinal image independent of pupil size, the transmission of high-frequency contrast is enhanced by pupil constriction (Figures 5, 7, and 8). The band of spatial frequencies that most depend on the pupil differs between mice (0.5–2 cpd) and humans (2–20 cpd), coinciding with high-acuity vision in both species.[89,110,112,113] Thus, our comparative computational modeling supports the hypothesis, first stated by Denton[114] and elegantly elaborated by Laughlin,[13] that the pupil maximizes acuity across a wide luminance range.

Combining cell-type-specific neuron removal and pharmacological activation of the sphincter pupillae (i.e., the muscle constricting the pupil), we show that the pupil bidirectionally controls the contrast sensitivity and acuity of the gaze-stabilizing optokinetic reflex in mice (Figure 6). We also discovered that pupil dilation disrupts prey detection and pursuit in mice (Figure 6). Predation depends on acuity and likely drove the evolution of retinal acute zones, including the primate fovea.[91–93,115,116] Studies with artificial pupils in humans found that smaller pupils enhance acuity in conscious visual perception.[12,14] Together, these studies indicate that the pupil's impact on high-acuity vision is conserved from mice to humans and across a wide range of behaviors.

Interactions of the pupil and the gaze

The pupil is one component of active vision; the other is the gaze. Combined with previous findings, our data suggest that the two components of active vision interact to optimize vision. First, the PCoR senses temporal contrast in a frequency band that overlaps with saccadic head and eye movements,

Figure 8. Human PCoR and retinal image

(A) Schematic diagram (i) of the behavioral apparatus used to assess pupillary light responses to centrally located stimuli in humans and the spectral distribution (ii) of the LCD stimulus (black) overlaid with the scaled spectra of all human photoreceptors (S-opsin, M-opsin, L-opsin, melanopsin, and rhodopsin).

(B) Average trace (±SEM) (i) at $10^{1.5}$ R* and summary data (ii) of the PLR (n = 3).

(C) Average trace (±SEM) (i) at $10^{1.5}$ R*, 50% contrast, 1 Hz, and summary data (ii) of the PCoR (n = 7).

(D) Dependence of the effective retinal illuminance (i.e., the retinal illuminance for cone vision, considering the SCE, color coded) on corneal illuminance and pupil size (i) and the relative (−SCE) and relative effective (+SCE) retinal illuminance as a function of pupil size alone (ii).

(E) Relative optical transfer (color coded) as a function of spatial frequency and pupil size (i), with direct comparisons of 1 and 10 cpd (ii) for M-opsin-weighted vision.

(F) Optical transfer as a function of pupil size (i) across the physiological range of human pupil radii from dilated (gray) to constricted (black) weighted for S-opsin. Experimentally determined pupil radii for an illuminance step (PLR, black) vs. contrast modulation at that illuminance (PCoR, green) exhibit significantly different curves ($p < 0.0001$ by bootstrapping).

(G) Analogous to (F) for M-opsin.

(H) Analogous to (F) for L-opsin.

(I) Analogous to (F) for rhodopsin.

(J) Analogous to (F) for melanopsin.

(K) Comparison of the retinal image for a dilated human pupil (i), illuminance constricted pupil (ii), and contrast constricted pupil (iii).

(L) Distributions of power across spatial frequencies in naturalistic movies (black traces) and their modulation by pupil constriction (i), fixational eye movements (ii), and the combination of pupil constriction and fixational eye movements (iii).

See also Figure S8.

translating spatial into temporal information.[4–7,9,96,117] The PCoR, in turn, measures the change in temporal contrast and proportionally enhances spatial contrast. Second, the spectral power of natural scenes falls with frequency (f) according to a power law ($\sim 1/f^2$).[118] Fixational eye movements decrease spectral power at low frequencies (Figure 8L),[9,119] whereas pupil constriction enhances spectral power at high frequencies (Figure 8L). Thus, the pupil and the gaze cooperate in whitening (i.e., decorrelating) the retinal input (Figure 8L) to reduce redundancies and improve the efficiency of information encoding in the retinal output.

A neural pathway for pupillary contrast and luminance responses

We identify a cell-type-specific pathway through the mouse retina that mediates the PCoR. This pathway starts in rod photoreceptors, whose phototransduction drives the PCoR (Figure 2). Rod signals can reach the inner retina via three pathways.[69] In the primary pathway, rods transmit signals to rod bipolar cells, which innervate AII amacrine cells, passing signals to the axons of cone bipolar cells.[120–122] In the secondary pathway, rods transmit signals through gap junctions to cones, which provide input to the dendrites of cone bipolar cells.[70,123,124] In the tertiary pathway, rods communicate directly with the dendrites of specific OFF cone bipolar cell types.[125–127] Removal of cone photoreceptors did not affect the PCoR, indicating that the secondary pathway is dispensable for signal propagation (Figure 3). Next, we found that type 6 ON cone bipolar cells, which do not directly contact rods, are required for the PCoR (Figure 3). This eliminates the tertiary pathway and suggests that rod signals for the PCoR reach the inner retina via the primary pathway.

Contrast information diverges into parallel channels at the level of cone bipolar cells, whose 14 types differ in their polarity (ON vs. OFF), spatial, temporal, and chromatic contrast preferences.[35,37,128,129] The PCoR dependence on type 6 bipolar cells identifies a first link between a specific cone bipolar cell type and a visual behavior. It will be interesting to map the contributions of other contrast channels and bipolar cell types to behavior. Using optogenetics, we show that type 6 bipolar cells are the primary source of synaptic excitation to M1 ipRGCs (Figure 3), consistent with anatomical reports of en passant synapses between type 6 axons and M1 dendrites.[31–33] Combined with the disruption of the PCoR in M-DTR mice (Figure 3) and the F1 → F0 conversion from synaptic excitation to spike output (Figure 4), this suggests that M1 ipRGCs convey the retinal output signals that drive the PCoR.

Type 6 bipolar cell ablation shifts the PLR to higher light levels that can drive pupil constriction through the intrinsic phototransduction of M1 ipRGCs (Figure 3). Thus, the same retinal pathway (rod → rod bipolar → AII amacrine → type 6 bipolar → M1 ipRGC) mediates the PCoR and the PLR.[16,17,42,43] However, whereas the PCoR relies entirely on contrast information encoded in the synaptic input to M1 ipRGCs, the PLR combines luminance signals from the synaptic input and intrinsic excitation. How the synaptic input from type 6 bipolar cells conveys luminance and contrast information[130] and whether the ultrastructure of en passant synapses, which cluster multiple presynaptic ribbons at a single synapse, contributes to this multiplexed transmission remains to be tested.

Thus, pupil responses to contrast and luminance are mediated by M1 ipRGCs, which multiplex signals for the PLR and the PCoR in their spike trains (Figure 4). Combining luminance and contrast measurements in the same pathway could help constrict the pupil to the stimulus-dependent maximum that maintains visibility, which depends on luminance and contrast. The specific architecture and function of the retinal M1 ipRGC circuit (Figures 2, 3, and 4) sets this balance. How this circuit and M1 ipRGC function vary across species to maintain balance and adjust it to species-specific goals remains to be explored. We observe pupil constriction at lower illuminances and temporal frequencies in diurnal humans compared with nocturnal mice (Figures 1 and 8), potentially reflecting species-specific optimizations for stimulus discrimination vs. detection.

Intrinsic excitability and the PCoR

In the PCoR/PLR pathway, phasic contrast signals are converted into tonic responses that drive pupil constriction by the M1 ipRGCs' transformation of excitatory input into spike output (Figure 4). The increased firing of M1 ipRGCs to phasic vs. tonic inputs could involve activation of I_h currents by the former[131] and spike frequency adaptation or depolarization block elicited by the latter.[38] Independent of the mechanism, it highlights the importance of intrinsic excitability to retinal computations[132] and provides the first behaviorally relevant example. The intrinsic excitability of M1 ipRGCs varies widely, producing a distributed population code for retinal illuminance.[29,38,66] Whether the same variability diversifies contrast encoding, and how population codes for luminance and contrast are multiplexed, remains to be determined.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.neuron.2024.04.012.

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AUTHOR CONTRIBUTIONS

M.J.F. and D.K. conceived of the project and wrote the manuscript. J.K. and M.J.F. performed predation experiments, which were analyzed by J.K., M.J.F., and D.K. J.-C.H. performed and analyzed two-photon calcium imaging experiments. N.S. carried out intraocular AAV injections and analyzed bipolar cell anatomy. M.J.F. performed and analyzed all other experiments.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-cone arrestin	Millipore	RRID: AB_1163387
Goat anti-cholineacetyltransferase	Millipore	RRID: AB_11213095
Chicken anti-GFP	ThermoFisher	RRID: AB_2534023
Mouse anti-RFP	Abcam	RRID: AB_1141717
Rabbit anti-melanopsin	Advanced Targeting Systems	RRID: AB_1608077
Mouse anti-SMI-32	BioLegend	RRID: AB_509997
Mouse anti-synaptotagmin 2	Zebrafish International Resource Center	RRID: AB_10013783
Mouse anti-calbindinD28k	Synaptic Systems	CAT# 214011; RRID: AB_2068201
Goat anti-OPN/Spp-1	Bio-Techne	CAT# AF1433
Donkey anti-chicken IgY Alexa 488	ThermoFisher	RRID: AB_2534096
Goat anti-mouse IgG Alexa 488	ThermoFisher	RRID: AB_2534084
Donkey anti-rabbit IgG Alexa 568	ThermoFisher	RRID: AB_2534017
Donkey anti-mouse IgG Alexa 568	ThermoFisher	RRID: AB_2534013
Donkey anti-goat IgG Alexa 633	ThermoFisher	RRID: AB_2535739
Bacterial and virus strains		
AAV5-Syn-hM4D(Gi)-mCherry	Bryan Roth	Addgene catalog # 50475
rAAV-GFP	Edward Boyden	Addgene catalog #37825
AAV1/2-Grm6-Cre	Johnson et al.[133]	N/A
Chemicals, peptides, and recombinant proteins		
Alexa Fluor 488 hydrazide, sodium salt	ThermoFisher	CAT# A10436
Alexa Fluor 568 hydrazide, sodium salt	ThermoFisher	CAT# A10437
Diphtheria toxin	Sigma	CAT# D0564
Cholera toxin subunit B, Alexa Fluor 488/594 conjugate	ThermoFisher	CAT# C34775/C34777
Retrobeads	Lumafluor	N/A
Vectashield	Vector Laboratories	RRID: AB_2336789
DAPI (4',6-Diamidino-2-Phenylindole, Dihydrochloride)	ThermoFisher	RRID: AB_2629482
Experimental models: Organisms/strains		
C57BL/6J	The Jackson Laboratory	Stock #000664
Gnat1 KO	The Jackson Laboratory	Stock #008811
$Gnat2^{cpfl3}$	The Jackson Laboratory	Stock #006795
Opn4-Cre	Dr. S Hattar	Ecker et al.[19]
Cck-Cre	The Jackson Laboratory	Stock #012706
Opsin-Cre	The Jackson Laboratory	Stock #032911
HiDTR	The Jackson Laboratory	Stock #007900
Ai9	The Jackson Laboratory	Stock #007909
Ai32	The Jackson Laboratory	Stock #024109
Ai148	The Jackson Laboratory	Stock #030328
Software and algorithms		
MATLAB	The Mathworks	RRID: SCR_001622
ISCAN	ISCAN	https://iscaninc.com/

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Python 3 (Anaconda distribution)	Anaconda	https://www.anaconda.com/
Cogent Graphics Toolbox	LON	http://www.vislab.ucl.ac.uk/cogent_graphics.php
PsychoPy	Peirce[134]	https://www.psychopy.org/
Fiji	Schindelin et al.[135]	RRID: SCR_002285
ISETBIO	Brainard et al.[85]	https://github.com/isetbio/isetbio
Original code	This paper	Zenodo:

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Daniel Kerschensteiner (kerschensteinerd@wustl.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- The original code has been deposited at https://github.com/kerschensteinerd/Fitzpatrick_2024 and is publicly available as of the date of publication. The DOI is listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

For this study, we utilized the following mouse lines: control wild-type (C57BL/6J, The Jackson Laboratory, stock #000664), Gnat1 KO[136,137] (Gnat1^{irdr}, The Jackson Laboratory, stock #008811), Gnat2^{cpfl3}[63] (The Jackson Laboratory, stock #006795), Opn4-Cre,[19] Cck-Cre[138] (The Jackson Laboratory, stock #012706), Opsin-Cre,[139] (The Jackson Laboratory, stock #032911), floxed diphtheria toxin receptor[140] (HiDTR, The Jackson Laboratory, stock #007900), floxed tdTomato[141] (Ai9, The Jackson Laboratory, stock #007909), floxed Channelrhodopsin-2[142] (Ai32, The Jackson Laboratory, stock #024109), and floxed GCaMP6f[77,78] (Ai148, The Jackson Laboratory, stock #030328). All procedures in this study were approved by the Animal Studies Committee of Washington University School of Medicine (Protocol # 20-0055 and 23-0116) and performed in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals. We observed no sex-specific differences in our results and, therefore, combined data from male and female mice.

We generated B6$_{CCK}$-DTR mice by crossing homozygous Cck-Cre mice with homozygous HiDTR mice, leading to diphtheria toxin receptor expression in all Cck-expressing cells. Adult (>P60) B6$_{CCK}$-DTR mice were anesthetized with a mixture of ketamine (0.1 mg/g body weight) and xylazine (0.01 mg/g body weight). They were then intraocularly injected with 2 μL total volume of 5 ng/μL diphtheria toxin (DT) in PBS. Intraocular DT injections did not affect gross retinal morphology, as measured by retinal layer thickness (Figure S3). Loss of B6 was confirmed by staining for synaptotagmin2 (Syt2) in the S5 IPL sublamina (Figures S4D–S4F). To express channelrhodopsin in type 6 bipolar cells (B6$_{CCK}$-ChR2 mice), we crossed homozygous Cck-Cre mice to homozygous Ai32 mice. We generated Cone-DTR and M-DTR mice by crossing, respectively, homozygous Opsin-Cre or Opn4-Cre mice with homozygous HiDTR mice. Similar genetic strategies have been described to ablate cones[71] and melanopsin-expressing cells.[143] Adult mice were injected intraperitoneally with DT in PBS (0.1 μg/g body weight for Cone-DTR; 0.02 μg/g body weight for M-DTR mice) every other day for a total of four injections. Intraperitoneal DT injections did not affect gross retinal morphology, as measured by retinal layer thickness (Figure S3). The loss of cones was assessed by staining for cone arrestin in the outer plexiform layer (Figures S4A–S4C). Loss of melanopsin-expressing cells (M1-M3 ipRGC) was confirmed by staining for melanopsin in the ganglion cell layer, and loss of M1 ipRGCs specifically was assessed by the loss of melanopsin-positive dendrites in the S1 IPL sublamina (Figures S4G–S4K). Loss of M4 ipRGCs was assessed by double staining for calbindin and Spp-1 in the ganglion cell layer[144] (Figure S4). For all behavioral experiments involving lines crossed to HiDTR, controls included a mixture of wild-type animals and DT injected Cre+ / DTR- or Cre- / DTR+ animals. We crossed homozygous Opn4-Cre mice to homozygous Ai9 mice (Opn4-tdT mice) to identify ipRGCs for electrophysiological recordings.

Cell-type-specific ablation strategies

To assess the circuit elements of the PCoR, we generated three mouse lines to ablate cones (Cone-DTR), type 6 bipolar cells ($B6_{CCK}$-DTR), and M1 ipRGCs (M-DTR), respectively. In each line, Cre-mediated recombination directs expression of the diphtheria toxin receptor to specific retinal neurons, followed by diphtheria toxin injections into adult animals to ablate the targeted cell population. Here, we discuss in more detail the transgenic targeting strategies, potential off-target effects, and their relevance to our findings.

Cone-DTR

The Opsin-Cre line, also known as HRGP-Cre, expresses Cre in cone photoreceptors with little to no off-target expression.[140] The same Cre line has previously been used to generate Cone-DTR mice.[71,72] Diphtheria toxin injections in these mice were shown to ablate M- and S-opsin-expressing cones without affecting rods or other retinal neurons.[71,72] Cone arrestin staining confirmed that diphtheria toxin injections in our Cone-DTR mice ablated ~90% of the cones (Figure S4). This did not significantly change the PLR or the PCoR (Figure 3). Similarly, we observed no differences in the PLR or the PCoR of $Gnat2^{cpfl3}$ mice, homozygous for a hypomorphic allele of the cone-specific α-transducin, vs. wild-type mice (Figure 2). Other visual behaviors and electrophysiological measures show deficits when ~50% of cones are removed in adult animals.[71,72,145] Thus, we conclude that the PCoR does not originate in cones. Our results from Gnat1 KO mice, lacking the rod-specific α-transducin, indicate that rods drive the PCoR (Figure 2). The PCoR preservation in the Cone-DTR mice (Figure 3) suggests that the secondary rod pathway is not required for signal propagation, consistent with recent findings for the PLR.[43] Whether the secondary rod pathway can support the PCoR in the absence of the primary rod pathway, as it can the PLR,[43] remains to be tested.

$B6_{CCK}$-DTR

Among bipolar cells, Cck (cholecystokinin) is selectively expressed in type 6 cells.[34] Indeed, type 6 bipolar cells comprise more than half of the Cck-expressing cells in the inner nuclear layer (INL).[146] We assessed the loss of type 6 bipolar cells following diphtheria toxin injection by staining for synaptotagmin 2 (Syt 2), which labels type 2 and type 6 bipolar cell axons in the OFF and ON sublaminae of the inner plexiform layer (IPL), respectively.[34,147] In diphtheria toxin-injected $B6_{CCK}$-DTR mice, Syt 2 labeling was absent in the ON and unchanged in the OFF sublamina (Figure S4), indicating that we removed type 6 but not type 2 bipolar cells.

Cck-Cre also targets two amacrine cell types: A17 and starburst amacrine cells.[146] For the full-field stimuli used in our study, M1 ipRGCs receive little inhibition from amacrine cells and lack surround antagonism.[22] Thus, the potential amacrine cell ablation in $B6_{CCK}$-DTR mice is unlikely to affect M1 ipRGC and pupil responses. Cck is expressed in transient suppressed-by-contrast and ON-OFF direction-selective ganglion cells.[34,146,148] The Cck-expressing ganglion cells project to the superior colliculus, dorsal and ventral lateral geniculate nucleus, and the nucleus of the optic tract[146,149] which do not control the pupil. We ruled out extensive off-target retinal cell ablation by measuring outer nuclear layer (ONL), INL, and IPL thickness, and the density of cells in the ganglion cell layer (GCL), which were indistinguishable between diphtheria toxin-injected $B6_{CCK}$-DTR and control mice (Figure S3). Our intraocular diphtheria toxin injections do not reach Cck-expressing cells outside the eye.

Given these considerations, the pattern of retinal cell removal we document (Figures 3 and S3), our optogenetic observation that type 6 bipolar cells dominate synaptic excitation of M1 ipRGCs (Figure 3), and connectomics analyses,[150] we conclude that diphtheria toxin injections in $B6_{CCK}$-DTR mice affect the PLR and PCoR by removing type 6 bipolar cells.

M-DTR

There are six types (M1-M6) of ipRGCs in mice that express melanopsin (Opn4) in a decreasing gradient from M1 to M6.[19–24] The extent to which Opn4-Cre targets ipRGCs matches their melanopsin expression.[19] Therefore, we expected diphtheria toxin injections in M-DTR mice to preferentially ablate M1-M3 ipRGCs. Indeed, we found near complete ablation of M1-M3 and only partial ablation of M4 ipRGCs (Figure S4).

While ablation of M2-M6 ipRGCs could affect pupil responses, we believe the phenotype we observe (i.e., the loss of PLR and PCoR, Figure 3) is due to the loss of M1 ipRGCs. M1 ipRGCs project to the shell of the olivary pretectal nucleus (OPN), where they innervate neurons projecting to the Edinger-Westphal nucleus, which controls the pupil.[73,74] Selective ablation of OPN shell-projecting ipRGCs is sufficient to eliminate the PLR.[18] In contrast, M2-M6 ipRGCs project to the OPN core,[19,73] whose contributions to pupil constriction are unclear.

We find that the F0 responses of the M1 ipRGCs precisely match the F0 tuning of the PCoR and that the M1 ipRGC firing rate correlates with pupil area for both the PLR and the PCoR (Figure 4). The same is not true for other ganglion cells, including M4 ipRGCs (Figure S6). Finally, M1 ipRGCs are unique among ipRGCs in receiving dominant synaptic input from type 6 compared to other bipolar cell types.[150] We show that type 6 bipolar cells provide synaptic excitation to M1 ipRGC and demonstrate that the loss of type 6 bipolar cells disrupts the PLR and PCoR (Figure 3). If other ipRGCs drove the PCoR, we might expect their remaining bipolar cell inputs (e.g., from types 7, 8, and 9) to compensate in $B6_{CCK}$-DTR mice partially. Yet, we observe complete elimination of the PCoR, suggesting a ganglion cell reliant on type 6 input (i.e., the M1 ipRGC) mediates the PCoR.

Whereas M1 ipRGCs are the dominant retinal input to the OPN shell, most input to the OPN core comes from M6 ipRGCs.[76] To assess whether M6 ipRGCs were affected in M-DTR mice vs. control mice, we injected retroAAV-GFP virus into the OPN (Figure S5). Thus, we found that M-DTR mice lack M1 ipRGC input to the OPN, but the M6 ipRGC input to the OPN remains largely intact (Figure S5). Further, by injecting anterograde, fluorescently conjugated CTB tracers into the eye, we show that most retinal input to the

OPN core remains in M-DTR mice (Figure S5). These findings suggest that M6 ipRGCs and other RGCs projecting to the OPN core[151] are less affected in M-DTR mice and support the conclusion that disruption of the OPN shell innervation by M1 ipRGCs, rather than the OPN core innervation by non-M1 ipRGCs, underlies our behavioral observations.

M4 ipRGCs (or sONα RGCs) provide strong input to the dorsolateral geniculate nucleus of the thalamus for image-forming vision.[19,45,152] Diphtheria toxin-injected M-DTR mice performed normally in a visual cliff test (Figure S4), which requires intact retino-geniculo-cortical vision. Additionally, our ability to restore optokinetic responses (OKR) in M-DTR mice with carbachol eye drops (Figure 6) demonstrates that OKR deficits in M-DTR mice are due to increased pupil size alone. Furthermore, dilating the pupils of control mice with atropine eye drops phenocopies the OKR deficits of M-DTR mice in the OKR and predation (Figure 6). Together, these observations indicate that the behavioral deficits we observe in M-DTR mice are due to increased pupil size and not any unintended effects on the retina.

Humans

The study included seven participants, ages 25-48, two of whom were authors (M.J.F. and D.K.) and five others not affiliated with the authors in any way (five males and two females). All participants had normal vision (with or without correction) and provided informed consent. The Institutional Review Board of Washington University School of Medicine (IRB ID# 202205098) approved all experimental procedures.

METHOD DETAILS

Animal surgeries

For all pupillometry and optokinetic experiments, mice were headplated under isoflurane anesthesia approximately one week before behavioral testing. Headplates were secured directly to the skull using Vetbond (3M) and dental cement (Parkell). We gave mice a minimum of 3 days to recover before acclimating to the appropriate behavioral apparatus and head-fixed holder. The acclimation duration gradually increased each day until the expected duration of experiments was reached.

For all brain injections, mice were anesthetized under isoflurane anesthesia. To label M1 ipRGCs, red fluorescent retrobeads (Lumafluor, Ex = 530 nm, Em = 590 nm) were injected with a Nanoject II (Drummond) bilaterally to SCN (± 0.15 mm M/L; - 0.46 mm A/P; 5.65 mm depth) of $B6_{CCK}$-ChR mice via a single midline craniotomy or bilaterally to OPN (± 0.62 mm M/L; 2.54 mm A/P; 2.19 mm depth) of Opn4-tdT mice via two craniotomies over either hemisphere. Electrophysiology experiments were performed 48-72 h after retrobead injection. For prey capture experiments, a virus constitutively expressing inhibitory DREADDs (AAV-DREADDi) was injected bilaterally into the OPN of wild-type mice via two craniotomies over either hemisphere. The AAV-DREADDi virus (AAV5-Syn-hM4D(Gi)-mCherry) was a gift from Bryan Roth (Addgene #50475). For retrograde labeling experiments of OPN projecting RGCs, a retro-AAV expressing GFP (rAAV-GFP, Addgene #37825) was injected bilaterally into OPN of wild-type and M-DTR mice. Virus injections were performed four weeks before experiments (i.e., at least three weeks before the onset of training for cricket hunting experiments). Injection quality was monitored by analyzing brain slices following the experiment's conclusion.

For AAV subretinal injections, mouse pups (postnatal days 4–6) were anesthetized on ice, and 200 nL AAV1/2-Grm6-Cre was delivered into the subretinal space using a Nanoject II injector (Drummond).

For intraocular injections, adult mice were anesthetized with ketamine (0.1 mg / g body weight) and xylazine (0.01 mg/g body weight) mixture. Either 2 μL diphtheria toxin (DT, Sigma D0564, 5 ng/μL in PBS) or 500 nL cholera toxin subunit B (CTB) conjugated to Alexa Fluor 488 or 594 (ThermoFisher, C34775/C34777, 1 mg/mL in PBS) was delivered into the vitreous chamber bilaterally using a Nanoject II injector (Drummond).

Pupil responses and eye tracking in mice

Visual stimulation dome

To assess pupil responses to spatially patterned stimuli, we constructed a custom visual stimulation dome that covers ~180° azimuth and ~120° elevation of the mouse visual field. The dome was built from a clear 18-inch diameter poly-carbonate sphere (California Quality Plastics) and coated with custom UV-reflective paint (Twilight Laboratory) designed to evenly reflect spectra from 350–600 nm while minimizing interference from internal reflections.[153] Mice were head-fixed on top of a cylindrical treadmill at the center of the dome, allowing the mouse to run freely while the eyes were positioned at the center of the spherical coordinate system created by the dome. Running movement on the treadmill was translated by a rotary encoder (Yumo) and encoded by an Arduino. Hot mirrors (Edmund Optics) were laterally mounted next to each eye, transmitting visible light from the dome while reflecting infrared light to an infrared-sensitive monochromatic CMOS camera (15 fps; ThorLabs). Infrared LEDs (ThorLabs) in the spherical dome arena allowed the cameras to capture pupil responses both during darkness and during visual stimuli. Videos were analyzed with DeepLabCut[154,155] by using eight points around the circumference of the pupil and fitting an ellipse.

Stimuli were projected onto the dome's surface via an E4500 MKII PLUS II projector (EKB Technologies) illuminated by 385, 460, and 520 nm LEDs to match the complement of mouse photoreceptors and reflected off a spherical mirror (Grainger). All stimuli were presented using PsychoPy[134] and warped to the dome's dimensions using a modified version of the meshmapper utility (http://paulbourke.net/dome/meshmapper/). The warp calibration was fine-tuned to correct spatial distortions across the dome from the perspective of the mouse's visual field. The projector was gamma-corrected and calibrated to ensure linearity of illuminance values, and ND filters (ThorLabs) were placed in the projector light path to control the dome illuminance.

To assess the PLR, illuminance steps (0.14–4.14 rhodopsin isomerizations/rod/ s or R* in 1-$\log_{10}$ increments) were presented globally. Each stimulus presentation comprised 5 s of background darkness, 30 s of illumination, and 30 s of post-illumination recovery to baseline, with a 2-min minimum of darkness between presentations. Pupil constriction was normalized to the dark-adapted pupil area, and the relative pupil area for each illuminance was calculated as the 5 s average around the maximum pupil constriction. To derive EC_{50} values, a Hill equation was fit to the data for each animal.

To assess the PCoR, the mouse was first light adapted for 10 min at 3.1 $\log_{10}$ R*. Then, sine-wave checkerboards of various spatial (0.001, 0.003, 0.01, 0.03, 0.1 cycles per degree or cpd) frequencies at 50% contrast were inverted at different temporal (0, 0.1, 0.2, 0.5, 1, 2, 5, 10, 30 Hz) frequencies. Each stimulus presentation comprised 5 s of steady-state illuminance, 30 s of sign-inversions, and 30 s of post-stimulus recovery, with a 30-s minimum of steady illuminance between presentations. Pupil constriction was normalized to the light-adapted pupil area, and the relative pupil area for each illuminance was calculated as the 5 s average around the maximum pupil constriction (F0 response). Traces were Fourier transformed to calculate the power at the stimulating frequency (F1 response), which was log normalized to the maximum response. To remove the effects of locomotion and arousal on pupil dilation, stimulus repeats in which locomotion exceeded 0.5 cm/s for >2 s or the pupil dilated >50% above baseline were excluded from the analysis.

Eye tracking

In other circumstances, pupil responses were assessed as previously described.[50] For pupillometry experiments, pupil size was tracked and recorded from the left eye using an ETL-200 eye-tracking system (ISCAN) under infrared illumination. At the same time, stimuli were presented to the mouse's right eye using an Arduino-controlled 525 nm LED and a set of ND filters (Thorlabs). To assess the PLR, varying illuminance steps (0.0 - 5.0 $\log_{10}$ R* in 0.5-$\log_{10}$ increments) were utilized. To assess the PCoR, the LED was square-wave modulated at various temporal frequencies (0, 0.1, 0.2, 0.5, 1, 2, 5, 10, 30 Hz) and contrasts (1, 5, 10, 25, 50, 75, 100%) after 10 min of light adaptation at 3.5 $\log_{10}$ R*. Analysis of PLR and PCoR data acquired on the ETL-200 was identical to data acquired on the dome. In some experiments, the background light intensity was raised to 5 $\log_{10}$ R* or lowered to 2.5 $\log_{10}$ R*.

For OKR experiments, visual stimuli were presented using the Cogent Graphics toolbox (John Romaya, Laboratory of Neurobiology at the Wellcome Department of Imaging Neuroscience, University College London) in MATLAB (The MathWorks). Square wave gratings of varying spatial frequencies (0.05, 0.067, 0.1, 0.13, 0.2 cpd) and Michelson contrasts (5%, 7.5%, 10%, 15%, 25%, 100%) moving at 10°/s in the temporal-to-nasal direction were presented on a monitor 16 cm from the mouse's left eye at a 45° angle. Each stimulus presentation comprised 10 s of a uniform gray screen, 60 s of drifting gratings, and a final 10 s of a uniform gray screen. Eye-tracking movements were quantified as the number of saccades preceded by a slow tracking motion. Wild-type animals received either PBS or 1% atropine sulfate eye drops. For M-DTR experiments, 100 mM carbamylcholine chloride in PBS was applied in 10 μL drops to both eyes for 5 min before being washed off in PBS, or PBS alone was applied. For OKR experiments in animals with dilated pupils, we found that the ISCAN system failed to capture eye movement accurately. Therefore, for all OKR experiments, raw video from the ETL-200 eye tracker was directly processed using DeepLabCut.

Prey capture

Cricket hunting experiments were performed as previously described.[48,50] Briefly, mice were individually housed and acclimated to the presence of the crickets 48 h before training. Food pellet deprivation began 16–18 h before training, and three crickets were provided to each mouse. Mice were acclimated to the arena on the first day of training, after which a cricket was placed in the center of the arena while recording hunting interactions with an overhead camera (30 fps; C310, Logitech). Mice had a maximum of 5 min to capture prey. Prey capture training continued for 4 days with three trials each day. After each day, mice returned to their home cages, were given access to food pellets for 6–8 h, and were then deprived of food pellets and given three crickets. Mice were trained for 5 days until their hunting performance plateaued, at which point they entered the testing phase of the behavior.

In experiments involving eye drops, animals were acclimated with PBS eye drops on training days 4 and 5. On test day, the same mice received both PBS and 1% atropine sulfate eye drops and were given between three to five opportunities to capture crickets. PBS eye drops always preceded atropine eye drops due to the prolonged duration of atropine-mediated pupil dilation in mice,[156] and trial sets were spaced >4 h apart to minimize day-to-day variability within mice.

Mice expressing DREADDi from bilateral AAV injections into OPN received PBS intraperitoneal (i.p.) injections on training days four and five to acclimate to the injection. The same mice received both PBS and CNO (1 mg/kg) i.p. injections on the test day. For PBS- and CNO-injected trials, mice had four opportunities to capture crickets. All behavioral results are reported from CNO- and PBS-injected trials spaced >4 h apart on test day to minimize day-to-day variability within mice.

Videos were analyzed in DeepLabCut[155,156] to track the position of the cricket and the mouse's ears, nose, and tail base. Custom software (OpenCV, Python) was used to extrapolate the mouse's position relative to the cricket, as previously described.[50] For each trial, we calculated the overall time to capture, the number of approaches made by the mouse, and the interval between approaches. Approaches were defined as periods when the mouse was running at speeds > 10 cm/s and decreased the distance between the mouse and the cricket by >7 cm/s. Approaches concluded when these criteria were no longer met or when the mouse contacted (distance < 4 cm) the cricket.

Analysis of eye position and pupil size during cricket hunting behavior was performed on a previously published dataset.[6] Probability of ocular vergence values during approach vs non-approach periods was calculated across each trial by binning vergence values into 10° bins, while probability of the normalized pupil radius was calculated with 5 percentile bins. To plot heatmaps of ocular vergence and mouse speed as a function of pupil radii, finer bins of 1°, 0.5 cm/s, and 1 percentile were used, and values were equally weighted per mouse (n = 7).

Visual cliff

Mice were placed on a central ridge (3.8 cm height x 1.7 cm width) across a 56 cm wide x 41 cm deep clear Plexiglas platform. One side of the ridge displayed a checkerboard pattern immediately underneath the platform (i.e., the shallow side), while the other side displayed an identical checkerboard pattern 61 cm beneath the platform (i.e., the deep side). Preference for stepping to the shallow versus deep side was measured across ten trials, with the relative orientation of the shallow and deep side randomly alternating between trials.

Tissue preparation

Mice were deeply anesthetized with CO_2, killed by cervical dislocation, and enucleated. For two-photon calcium imaging and electrophysiology experiments, mice were dark adapted for >2 h before euthanization. Retinas were isolated under infrared illumination (>900 nM) in oxygenated mouse artificial cerebrospinal fluid (mACSF-$NaHCO_3$) containing (in mM) 125 NaCl, 2.5 KCl, 1 $MgCl_2$, 1.25 NaH_2PO_4, 2 $CaCl_2$, 20 glucose, 26 $NaHCO_3$, and 0.5 L-glutamine equilibrated with 95% O_2 and 5% CO_2 and flat mounted on a membrane disc (Anodisc, Whatman).

For immunohistochemical experiments, retinas were isolated in oxygenated mACSF with HEPES buffer (mACSF-HEPES) containing (in mM) 119 NaCl, 2.5 KCl, 1.3 $MgCl_2$, 1 NaH_2PO_4, 2.5 $CaCl_2$, 11 glucose, and 20 HEPES, with pH adjusted to 7.37 using NaOH and either flat mounted on black membrane disks (Millipore Sigma) or embedded in 4% agarose to cut 60 μm vertical slices close to the optic nerve. In some cases, brains were isolated and embedded in 4% agarose to cut 100 μm coronal sections centered on the OPN.

Immunohistochemistry

After isolation or two-photon calcium imaging, retinas were transferred to membrane disks (HABGO1300, Millipore) and fixed with 4% paraformaldehyde for 30 min in mACSF-HEPES. Isolated brains were drop-fixed in 4% paraformaldehyde in PBS for 4 h at RT and then overnight at 4°C. For immunostaining, whole-mounted retinas were cryoprotected in 10% sucrose in PBS for 1 h at RT, 20% sucrose in PBS for 1 h at RT, and 30% sucrose in PBS overnight at 4°C. After three freeze-thaw cycles, retinas were washed three times for 10 min in PBS at RT, blocked in 10% normal donkey serum (NDS) in PBS for 2 h at RT, and incubated with primary antibodies in 5% NDS and 0.5% Triton X-100 in PBS for 5 days at 4°C. Brain and retina slices were not cryoprotected but were blocked in 10% normal donkey serum (NDS) in PBS for 2 h at RT, and incubated with primary antibodies in 5% NDS and 0.5% Triton X-100 in PBS overnight at 4°C. We used the following primary antibodies in this study: rabbit anti-cone arrestin (CAR, 1:1000, Millipore, RRID:AB_1163387), goat anti-cholineacetyltransferase (ChAT, 1:200, Millipore, RRID:AB_11213095), chicken anti-GFP (1:1000, ThermoFisher, RRID:AB_2534023), mouse anti-RFP (1:1000, Abcam, RRID:AB_1141717), rabbit anti-melanopsin (UF006, 1:2000, Advanced Targeting Systems, RRID:AB_1608077), mouse anti-SMI-32 (1:500, BioLegend, RRID:AB_509997), mouse anti-synaptotagmin 2 (znp-1, 1:1000, Zebrafish International Resource Center, RRID: AB_10013783) mouse anti-calbindinD28K (1:2000, Synaptic Systems, Cat# 214011), goat anti-OPN/Spp-1 (1:40, Bio-Techne, Cat# AF1433). After incubation with primary antibodies, whole-mounted retinas were washed in PBS three times for 10 min at RT, and incubated with DAPI (300 nM) and secondary antibodies conjugated with Alexa 488 (1:1000, ThermoFisher, anti-chicken IgY RRID:AB_2534096; 1:1000, ThermoFisher, anti-mouse IgG RRID:AB_2534084), Alexa 568 (1:1000, Thermo-Fisher, anti-rabbit IgG RRID:AB_2534017; 1:1000, Thermo-Fisher, anti-mouse IgG RRID:AB_2534013), and Alexa 633 (1:1000, ThermoFisher, anti-goat IgG RRID:AB_2535739) overnight at 4°C and mounted in Vectashield medium (Vector Laboratories, RRID:AB_2336789) after three washes in PBS for 10 min at RT. Brain and retina slices were washed in PBS three times for 10 min at RT, incubated in appropriate secondary antibodies and DAPI for 2 h at RT, washed in PBS three times for 10 min at RT, and then mounted in Vectashield.

Confocal imaging and analysis

Confocal images were acquired on an Fv1000 laser scanning microscope (LSM, Olympus) using either a 60X 1.35 NA oil immersion objective or a 20X 0.85 NA oil objective. To assess the effect of DT injections in $B6_{CCK}$-DTR and M-DTR mice, images of multiple retinal slices or whole-mounts were taken to measure the thickness or densityof each layer. To assess cell-type-specific deletion in the Cone-DTR and M-DTR mouse lines, images were taken in multiple regions of the retina to compute cell density (cell / mm^2) of cone-arrestin-, melanopsin-, and calbindin/Spp-1-positive cells. For $B6_{CCK}$-DTR mice, z-stacks were acquired at multiple regions of the retina. The fluorescent intensity of synaptotagmin 2 was calculated at each z-depth through the IPL along with the DAPI signal, which was used to reference the ganglion cell and inner nuclear layers across mice. Following two-photon calcium imaging, we identified type 6 bipolar cells based on the stratification of their axon arbors below the ChAT bands in the IPL as well as dendritic territory (112.55 ± 13.71 μm^2), the numbers of cones contacted (3 ± 0.71), and axonal territory (144.54 ± 8.22 μm^2) as previously reported.[67,71,157,158] To determine GFP-labeled ganglion cells in rAAV-GFP injected mice, z-stack images were taken in multiple regions of the retina to capture both somata and dendritic trees so as to facilitate cell-typing. OPN images were acquired at the rostral head of the nucleus.

Two-photon calcium imaging

Image acquisition

We used a custom-built upright two-photon microscope from Scientifica, controlled by the Scanimage r3.8 MATLAB toolbox. Images were obtained using a DAQ NI PCI6110 data acquisition board from National Instruments. The imaging process involved exciting GCaMP6f using a Mai-Tai laser tuned to 930 nm, and fluorescence emission was captured using a 60X 1.0 NA water immersion objective from Olympus. The fluorescence signal was filtered through consecutive 450 nm long-pass filters from Thorlabs and 513–528 nm bandpass filters from Chroma to block visual stimulus light (peak: 385 nm) from reaching the PMT. GCaMP6f signals were captured at a scan rate of 37.9 Hz and an average pixel density of 0.27 pixels/pixels/μm^2 (min: 0.21 pixels/μm^2, max: 0.32 pixels/μm^2). To avoid recording near potential retinal detachments from the subretinal injections, the injection center was located, and isolated bipolar were recorded 0.5 – 1 mm away from the injection site. Before presenting each visual stimulus, the retina was adapted to the laser light for 30 s. All images were obtained from the ventral retina, where S-opsin dominates. During the experiments, the retinas were perfused at a rate of approximately 6 mL/min with 33 °C mACSF-$NaHCO_3$ equilibrated with 95% O_2 and 5% CO_2.

Image processing

We simultaneously acquired transmitted light and fluorescence images to detect z-axis displacements, which led to the rejection of the respective image series. Series without z-axis displacements were registered to the middle frame using built-in MATLAB functions, and rigid transformations were applied to both transmitted and fluorescence images. Registration quality was confirmed by visual inspection, and the transformed fluorescence images were subsequently processed and analyzed. Raw images underwent a 3D median filter (3 x 3 x 3 pixels) to remove shot noise. Standard deviation over the time domain of the recording estimated the quality of each pixel. Quality thresholds were manually adjusted by visual inspection to isolate signals from bipolar cell terminals and eliminate background noise. Stimulus responses of qualifying pixels were analyzed after subtraction of the average signal 0.5 s before the stimulus onset. Repeat reliability was calculated for each cell based on the correlation coefficient between repeated trials of the same stimulus condition. The threshold of median R^2 greater than 10% did not exclude any bipolar cells from the six retinae studied. Finally, for each bipolar cell, responses were normalized to the maximum response across all stimulus frequencies.

Visual stimulation and response analysis

The Cogent Graphics toolbox in MATLAB (The Mathworks) was used to generate visual stimuli in this study. The stimuli were presented via a UV E4500 MKII PLUS II projector, illuminated by a 385-nm LED (EKB Technologies), and focused onto the photoreceptors of the ventral retina using the substage condenser of the upright two-photon microscope (Scientifica). The average intensity of all stimuli was maintained at ~600 S-opsin isomerizations/cone/s. Stimuli were centered on the two-photon scan field. To evaluate frequency responses, a flickering 150-μm-diameter spot was presented with sinusoidal intensity modulation ranging from -50% to 50% at six different frequencies (0.5, 1, 2, 4, 8, 16 Hz). An 800 μm^2 gray background circle was presented throughout the experiment. Each frequency was repeated for 2 s, followed by a 1.5-s interval with gray background intensity, and this sequence was repeated five times with a random order of the six frequencies for each repeat. The F0 mean response was obtained by averaging the response of each frequency from stimulus onset to 2.5 s later, over five repetitions. The log-normalized F1 frequency power response was calculated using a Fourier transform on traces upsampled to 189.4 Hz using the MATLAB 1D shape-preserving piecewise cubic interpolation function.

To correct the frequency responses arising from the intrinsic properties of GCaMP6f, we simulated the corresponding frequency powers (F1) by convoluting the calcium response function[159] with signal amplitudes of equal magnitude. The function is defined as follows:

$$f(t) = A\frac{e^{\frac{-t}{\tau_r}} - e^{\frac{-t}{\tau_d}}}{\tau_r - \tau_d}$$

where A is an adjustable constant, t represents time, and τ_r and τ_d denote the rising and decay time.[160] Adjusted to match our recording temperature,[161] $\tau_r = 49.5\ ms$ and $\tau_d = 156.2\ ms$.

To accurately align with our recording data, we first sampled the ideal sine function at 1,000 Hz for 4 s as our stimulation time, which includes 1 s before stimulus onset and 1 s after 2 s of stimulus presentation. Subsequently, we created a GCaMP6f filter for 2 s. We then downsampled to match our two-photon recording frequency (i.e., 37.88 Hz) and aligned it with the signal time. Following this, we computed the power as described above. We normalized the powers of same-level simulated indicator signals to the maximum power, in the same manner as the experimental data. The corrected F1 powers were obtained as quotients, and the results were displayed on a $\log_{10}$ scale.

Electrophysiology

Patch-clamp recordings

Mice were dark adapted for >2 h prior to euthanization with CO_2, decapitation, and enucleation. Retinas were then isolated and, for the duration of the experiment, were constantly superfused with warm (30-35°C) mACSF-$NaHCO_3$ M1 ipRGCs were targeted for whole-cell patch-clamp recordings under two-photon microscopy (Fv-1000 MPE, Olympus; Mai-Tai HP, SpectraPhysics) either by retrograde fluorescent beads from SCN or OPN injections or by tdTomato fluorescence in Opn4-tdT mice. Correct targeting was confirmed morphologically by the inclusion of Alexa 488 or Alexa 568 (0.1 mM) in the patch pipette solution and subsequent two-photon imaging. For current-clamp and cell-attached recordings, the patch pipette solution contained (in mM) 125 K-gluconate, 10 NaCl, 1 $MgCl_2$, 10 EGTA, 5 HEPES, 5 ATP-Na_2, and 0.1 GTP-Na (pH adjusted to 7.2 with KOH); for voltage-clamp recordings, the patch pipette solution contained (in mM) 120 Cs-gluconate, 1 $CaCl_2$, 1 $MgCl_2$, 10 Na-HEPES, 11 EGTA, 10 TEA-Cl, 2 Qx314, ATP-Na_2, and 0.1 GTP-Na (pH adjusted to 7.2 with CsOH). Patch pipettes were pulled to resistances of 5-8 MΩ. Signals were amplified with a Multiclamp 700B amplifier (Molecular Devices), filtered at 3 kHz (eight-pole Bessel low-pass), and digitally sampled at 10 kHz (Digidata 1440A, Molecular Devices). In some cases, slow (τ = 5 s) current injection was performed in current-clamp mode to maintain a resting membrane potential of -60 mV and avoid depolarization block.[38] Excitatory postsynaptic currents (EPSCs) were measured at -60 mV. Series resistance was compensated by bridge balance (~15 MΩ) in current-clamp and electronically by ~75% in voltage-clamp recordings. Liquid junction potentials (~14 mV) were corrected offline.

For optogenetic experiments, the following drugs were used to block endogenous photoreceptor responses: L-AP4 (20 μM, Tocris), ACET (10 μM, Tocris), and HEX (300 μM, Tocris). Channelrhodopsin responses were evoked with light from a mercury bulb (Olympus), which was band-pass filtered (426 – 446 nm, Chroma) and focused onto the RGC side of the retina (intensity: 3.15 x 10^{-4} W/mm^2) through a 20X 0.95 NA water immersion objective. A Uniblitz shutter controlled stimulus timing (Vincent Associates). In some experiments, stimulus responses were evoked without drug application in Channelrhodopsin-2-negative animals to elicit photoreceptor-driven inward currents for comparison.

Visual stimulation and response analysis

Stimuli were presented via an Arduino-controlled 525-nm LED, passed through a set of ND filters (Thorlabs), and focused onto the photoreceptor layer through the microscope condenser. Varying light steps (1.0–5.0 $\log_{10}$ R* in 1.0-log10 R* increments) were utilized to assess illuminance responses. Firing rates were averaged for 5 s around the peak firing rate following the illuminance step. To assess contrast responses, the LED was square-wave modulated at various temporal frequencies (0, 0.1, 0.2, 0.5, 1, 2, 5, 10, 30 Hz) at 50% contrast after 10 min of light adaptation at 3.5 $\log_{10}$ R*. In some cases, 20 pA of square-wave current was injected at a subset of those frequencies (0.2, 1, 5, 30 Hz). Spike trains were binned into 1-s bins for F0 spike rate analysis and into 1/60 s bins for F1 power analysis. The F0 response was calculated as the difference between the average response during the contrast response stimulus and the baseline firing rate divided by the baseline-subtracted maximum firing rate (Δ Hz/Hz, and the F1 response was calculated via a Fourier transformation of the spike train and log-normalized to the maximum response. EPSCs were low-pass filtered at 10 Hz and downsampled to 100 Hz. The F0 response was calculated as the difference between the average response during the contrast response stimulus and the baseline inward current divided by the baseline-subtracted maximum inward current (ΔpA/pA). The F1 response was calculated via a Fourier transformation of the spike train and log-normalized to the maximum response.

EPSCs from optogenetic data were low-pass filtered at 10 Hz, and maximum current magnitude was defined as the peak inward current during the 100-ms stimulus presentation subtracted from the baseline current 100 ms before the stimulus. The time to half maximum (TTHM) was calculated from the same low-pass-filtered trace as the time from stimulus onset to half the value of the peak inward current.

Computational modeling

All modeling was based on the ISETBio package.[84–86] For human calculations, we used default parameters. For mouse calculations, we obtained parameters from the literature.[162–164] Optical transfer functions (OTFs) were calculated assuming diffraction-limited optics across the visible wavelength spectrum and then weighted according to the relative sensitivity of each photoreceptor class. Before generating retinal images appropriate for the mouse, fisheye lens images were taken from a camera (CT9000, Crosstour) at ground level to simulate the mouse's visual world. Images were calibrated at 19.42 pixels/degree, and we exclusively used the monochromatic green-blue spectrum. Calculated OTFs were then interpolated across two spatial dimensions at the same spatial resolution of the images and then multiplied by the Fourier transform of the image. Taking the inverse Fourier transform of the product yielded the simulated retinal image. Images appropriate for the human visual world were taken from YouTube, as previously described,[165] calibrated at 24.15 pixels/degree, and made monochromatic from all color channels.

Retinal illuminance was calculated as a function of the pupil radius as follows:

$$I_r = I_e \tau \frac{R_{Pupil}^2}{2f^2}$$

Where I_r is the retinal illuminance, I_e is the environmental illuminance, τ is the transmittance of the ocular media, R_{pupil} is the radius of the pupil, and f is the focal length of the mouse eye. To estimate the Stiles-Crawford effect, the lens was modeled as a 2D Gaussian where light passage through the center of the lens was unaffected and became fractionally smaller towards the periphery. Spread of the Gaussian was calculated based on the full width at half maximum of either ground squirrel cone light directionality data (~12 degrees)[83] for mice or human Stiles-Crawford effect measurements (~10 degrees)[80–82] and then masked by the pupil size.

Natural scene statistics

To compare the effect of fixational eye movements and pupil diameter on the power spectrum of natural images, we analyzed a database of images with recorded eye movements of observers (DOVES).[166] Using only fixational eye movements, we generated movies by centering the image on gaze fixation and translating it in each frame. Static spatial power spectra were generated by summing the squared Fourier transform of each frame, while the dynamic component was generated by summing the squared Fourier transform of each non-zero temporal frequency of the overall spatiotemporal power spectra, as previously described.[167] Power spectra were then multiplied by the calculated, M-cone-weighted human OTF at the same spatial resolution of the images for either a dilated (4-mm radius) or constricted (1-mm radius) pupil.

Pupil responses in humans

Participants were seated in front of a gamma-corrected calibrated desktop monitor 50 cm from the screen. All participants had 20/20 vision with or without correction. Videos of the right eye were taken under infrared illumination with a USB infrared camera (30 fps; ELP 720p). Visual stimulation was controlled using PsychoPy.

To assess the PLR, illuminance steps (-1–1.5 $\log_{10}$ R* in 0.5-$\log_{10}$ increments) were presented from darkness. To assess the PCoR, participants were first light adapted for 10 min at 1.5 $\log_{10}$ R*. Then, the monitor was square-wave modulated at various temporal frequencies (0, 0.1, 0.2, 0.5, 1, 2, 5, 10, and 30 Hz) at 50% contrast. Stimuli were identical to those shown in mice. They comprised 5 s of either darkness (PLR) or steady-state illuminance (PCoR), 30 s of stimulus presentation, and 30 s of post-stimulus recovery, with either a 120-s minimum of darkness (PLR) or 30-s minimum of steady illuminance (PCoR) between presentations. Since PCoR experiments were performed at illuminances driving near-maximal pupil constriction, trials were excluded if the light-adapted pupil area was <3.1 mm^2. Pupil constriction was normalized to either the dark-adapted (PLR) or the light-adapted (PCoR) pupil area, and the relative pupil area for each illuminance was calculated as the 5 s average around the maximum pupil constriction.

QUANTIFICATION AND STATISTICAL ANALYSIS

Unless otherwise noted, all population data are reported as mean ± SEM, and n represents the number of animals or cells analyzed. The p values were calculated using n-way ANOVAs, Kruskal-Wallis tests, Student's t tests, Mann-Whitney U tests, Kolmogorov-Smirnov tests, or bootstrapping as appropriate and as specified in the figure legends to assess the statistical significance of observed differences. For multiple comparisons, p values were corrected using Tukey's or Dunn's procedure. No statistical methods were used to predetermine sample sizes.

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Supplemental information

A pupillary contrast response in mice and humans:

Neural mechanisms and visual functions

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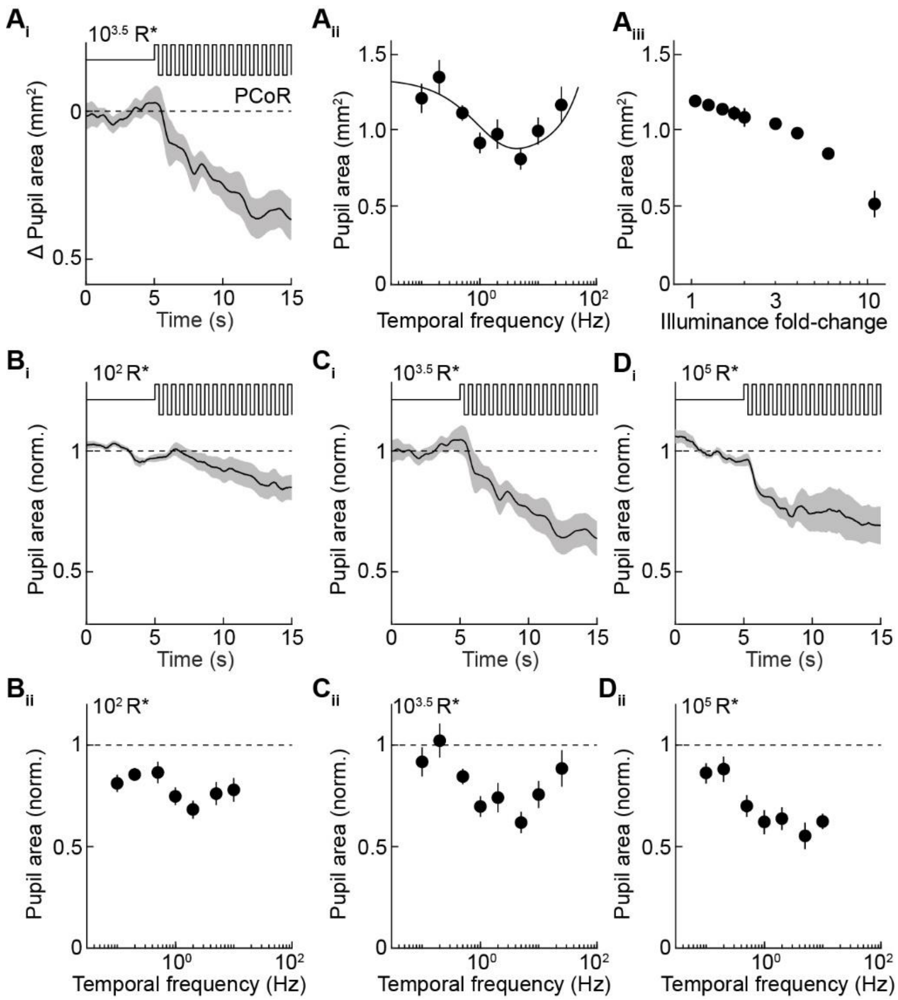

Figure S1. PCoR across different background illuminances. Related to Figure 1.

(A) Average trace (± SEM, i) and summary data (n = 22, ii) of pupillary contrast responses at $10^{3.5}$ R*. All traces in this figure show responses to 50% temporal contrast at 5 Hz. Comparison with the pupil area after an illuminance increase (iii) from the same $10^{3.5}$ R* background.

(B) Average trace (± SEM, i) and summary data (n = 12, ii) of pupillary contrast responses at 10^2 R*, normalized to the pre-stimulus pupil area.

(C) Analogous to B for $10^{3.5}$ R* (n = 22). The underlying data are the same as in A.

(D) Analogous to B for 10^5 R* (n = 13).

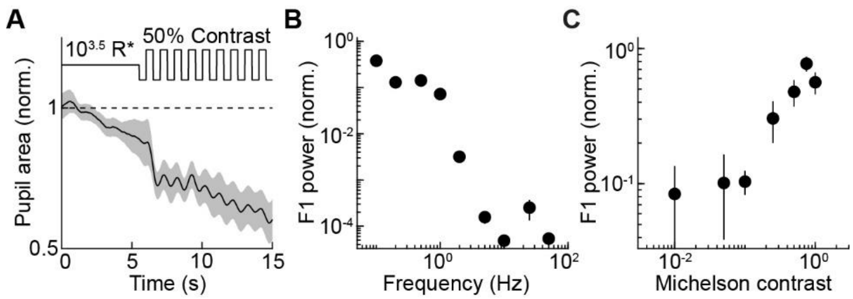

Figure S2. F1 responses of the pupil to temporal contrast. Related to Figure 1.

(A) Average responses trace (± SEM) of the pupil to a 50% temporal contrast at 1 Hz with $10^{3.5}$ R* mean luminance (n = 9).

(B) Summary data of the response power at different stimulus frequencies (i.e., the F1 responses) normalized to the maximal response (n = 15). All stimuli were presented at 50% contrast.

(C) Summary data of the F1 response power at different stimulus contrasts normalized to the maximal response (n = 9). All stimuli were presented at 1 Hz.

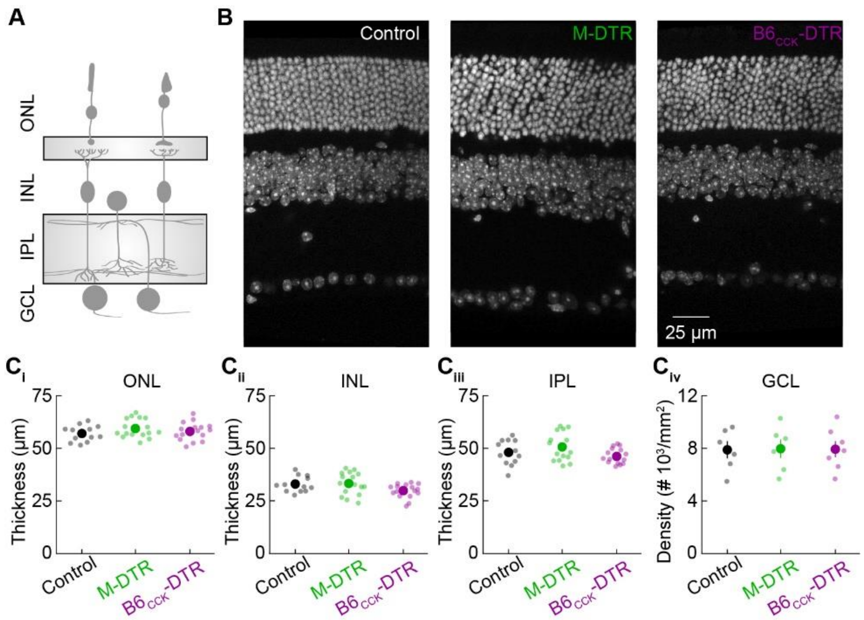

Figure S3. Retinal layer thickness following DTR-mediated cell ablation. Related to Figure 3.

(A) Schematic of the retina illustrating the location of the outer nuclear layer (ONL), inner nuclear layer (INL), inner plexiform layer (IPL), and ganglion cell layer (GCL).

(B) Representative images of DAPI staining of cell nuclei in vertical slices of control (left, n = 12 slices from 3 retinas), M-DTR (center, n = 16 slices from 4 retinas), and $B6_{CCK}$-DTR (right, n = 16 slices from 4 retinas) retinas.

(C) Summary data comparing ONL (i), INL (ii), and IPL (iii) layer thickness in vertical sections between control, M-DTR, and $B6_{CCK}$-DTR retinas ($p > 0.05$ by ANOVA). Summary data comparing GCL cellular density (iv) in whole-mounts between control (n = 6), M-DTR (n = 6), and $B6_{CCK}$-DTR (n = 7) retinas ($p > 0.05$ by ANOVA).

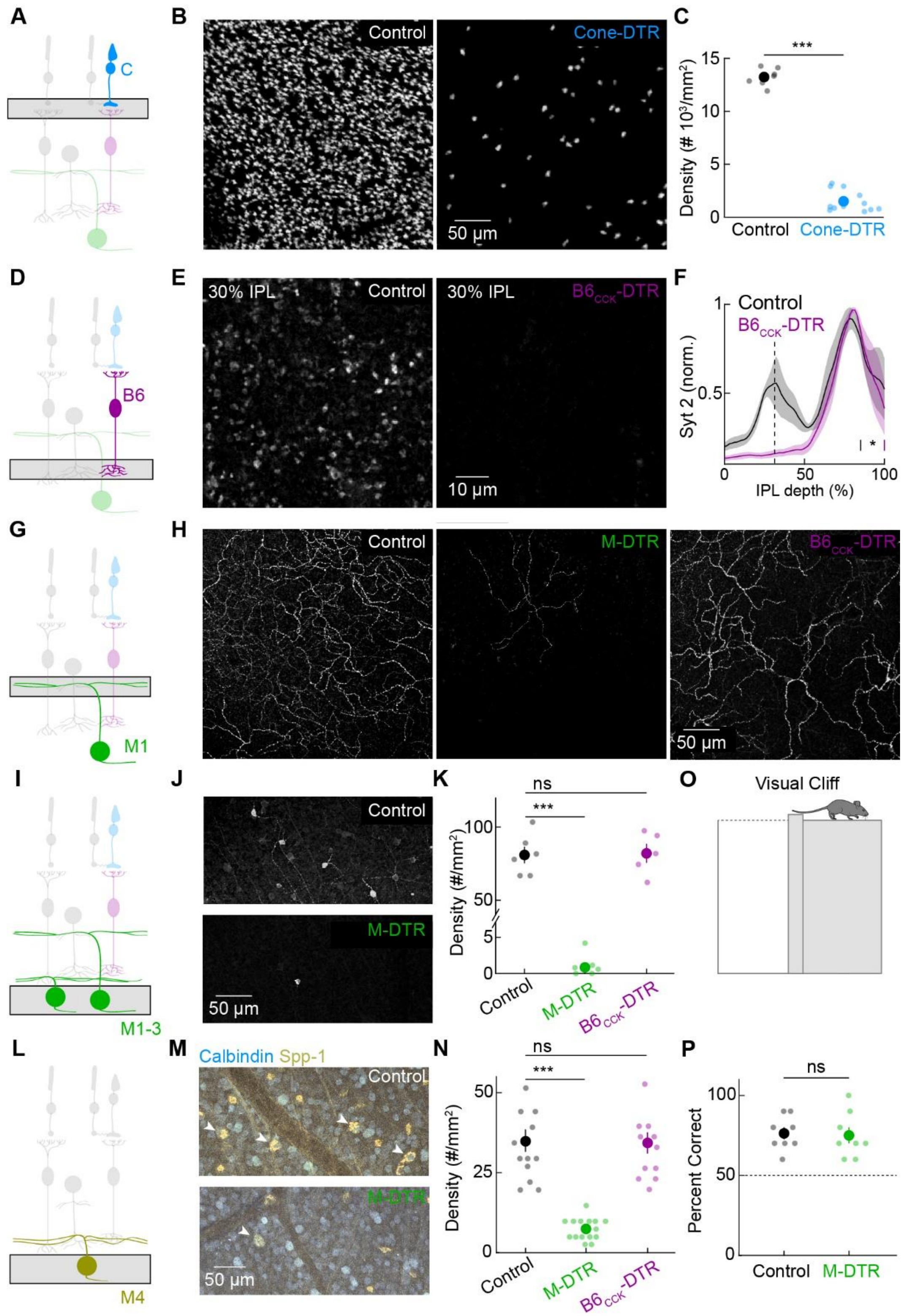

Figure S4. Anatomical confirmation of cell ablation approaches. Related to Figure 3.
(A) Schematic of the retina highlighting cones and the imaging plane for cone pedicles.

(B) Representative images of cone pedicles in the outer plexiform layer stained for cone arrestin in control (left) and Cone-DTR (right) retinas.
(C) Summary data comparing the cone density in control (n = 8 ROIs from 3 retinas) and Cone-DTR mice (n = 12 ROIs from 4 retinas, $p < 0.001$ by two-sample t-test).
(D) Schematic of the retina highlighting type 6 bipolar cells (B6) and the imaging plane for B6 synaptic terminals.
(E) Representative images of Syt2 staining in the inner IPL (30% IPL depth) of control (left) and $B6_{CCK}$-DTR retinas (right).
(F) Summary data showing the synaptotagmin 2 (Syt2) fluorescence intensity (mean ± SEM) across the depth of the inner plexiform layer (IPL). Syt2 labels type 2 and type 6 bipolar cells. The peak in the fluorescence intensity profile associated with type 6 bipolar cells (vertical dashed line) in the inner IPL is lost in $B6_{CCK}$-DTR mice (control: n = 4 retinas, $B6_{CCK}$-DTR: n = 4, $p < 0.001$ by two-sample t-test).
(G) Schematic of the retina highlighting M1 ipRGCs and the imaging plane for the M1 dendritic arbor.
(H) Representative images of M1 ipRGC dendrites in the IPL of control (left), M1-DTR (center), and $B6_{CCK}$-DTR retinas, as labeled by melanopsin staining.
(I) Schematic of the retina highlighting M1-M3 ipRGCs and the imaging plane, including their somas.
(J) Representative images of melanopsin staining in the GCL of control (top) and M-DTR (bottom) retinas. Melanopsin labels M1-M3 ipRGCs.
(K) Summary data comparing M1-M3 ipRGC density in control (n = 6 retinas), M-DTR (n = 6), and $B6_{CCK}$-DTR (n = 6) mice (control vs M-DTR $p<0.001$; control vs $B6_{CCK}$-DTR $p=0.99$ by ANOVA with Tukey's test).
(L) Schematic of the retina highlighting M4 ipRGCs (ON-sus-alphas) and the imaging plane, including their somas.
(M) Representative images of Calbindin and Spp-1 staining in the GCL of control (top) and M-DTR (bottom) retinas. Calbindin/Spp-1 double-positive cells are M4 ipRGCs.
(N) Summary data comparing M4 ipRGC density in control (n = 14 ROIs from 4 retinas), M-DTR (n = 15 ROIs from 4 retinas), and $B6_{CCK}$-DTR (n = 12 ROIs from 4 retinas) mice (control vs M-DTR $p<0.001$; control vs $B6_{CCK}$-DTR $p=0.98$ by ANOVA with Tukey's test).
(O) Schematic illustrating the experimental setup for testing visual cliff responses. Mice must correctly choose the shallow versus the deep side.
(P) Summary data comparing performance in the visual cliff test between control (n = 8 mice) and M-DTR (n = 8) mice ($p = 0.84$ by two-sample t-test). Dashed line indicates chance-level performance.

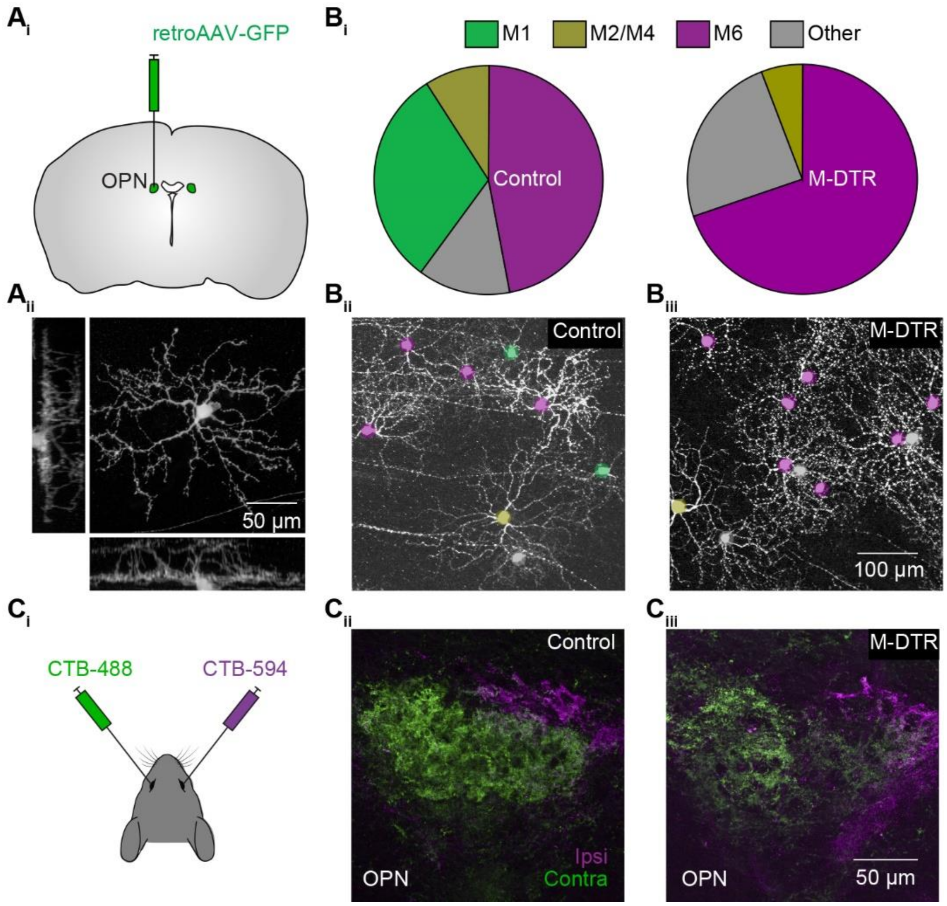

Figure S5. Retinal projections to OPN in M-DTR mice. Related to Figure 3.

(A) Schematic (i) illustrating the targeting of retro AAV-GFP virus injections to the OPN, and representative image (ii), with orthogonal views, of the most numerous labeled OPN-projecting cell in the retina, the M6 ipRGC.

(B) Distribution (i) of labeled retinal cell types (M1 ipRGCs, M2/M4 ipRGCs, M6 ipRGCs, or other) following retroAAV-GFP virus injections to the OPN in control (left, n = 68 cells from 4 retinas in 2 mice) and M-DTR (right, n = 49 cells from 6 retinas in 3 mice) mice. Representative images of GFP labeled RGCs in control (ii) and M-DTR (iii) mice with colored labels indicating each cell type.

(C) Schematic (i) of the intraocular injection strategy where different fluorescent conjugates of the tracer cholera toxin subunit B (CTB) are injected in the contralateral or ipsilateral eye. Representative images of the OPN showing contralateral and ipsilateral retinal input in control (ii) and M-DTR (iii) mice.

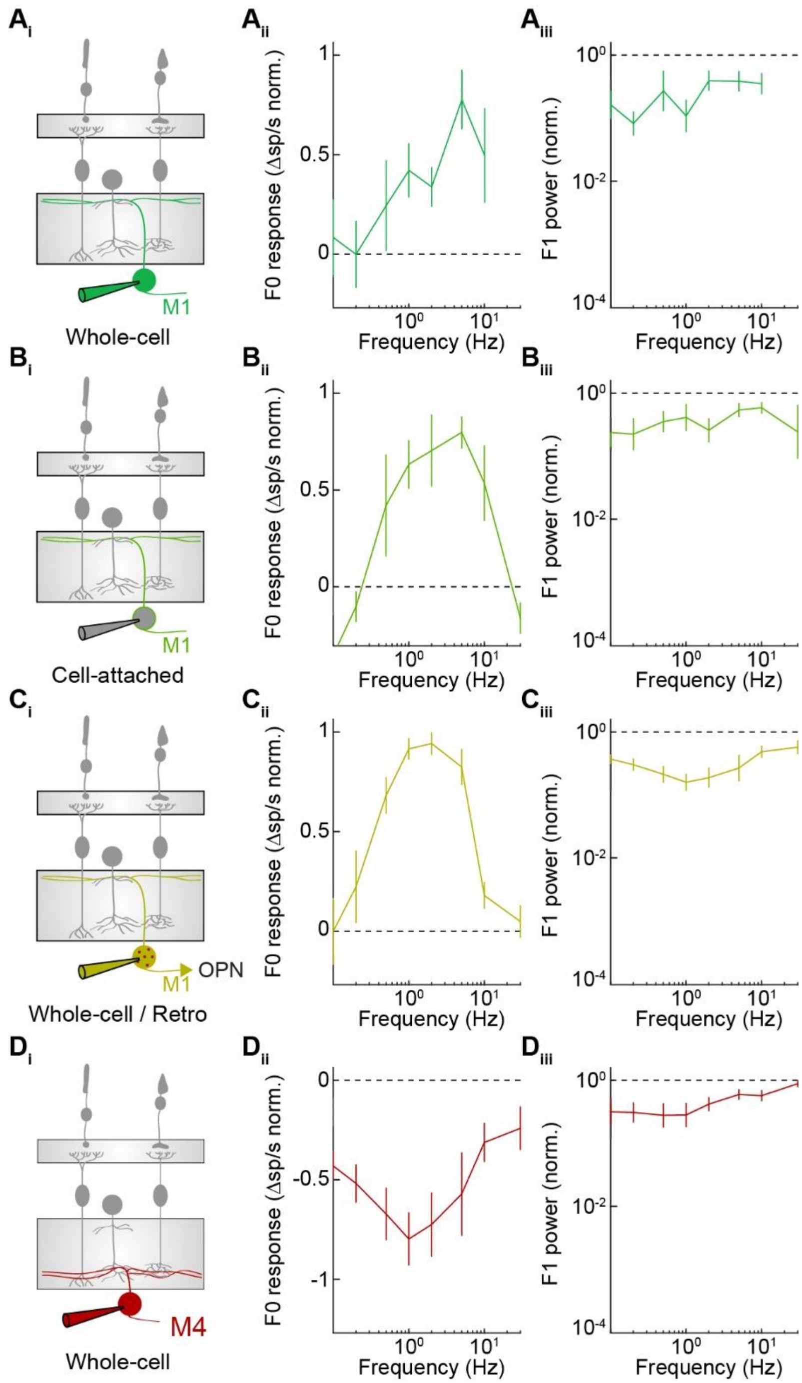

Figure S6. F0/F1 responses in M1 and M4 ipRGCs. Related to Figure 4.

(A) Whole-cell current-clamp recordings from M1 ipRGCs. Schematic of the retina with targeted cell type and recording strategy (i). F0 (ii) and F1 (ii) responses at each tested frequency (n = 6).

(B) Analogous to A, but for cell-attached recordings from M1 ipRGCs (n = 5).

(C) Analogous to A, but for whole-cell current-clamp recordings from retrogradely labeled, OPN-projecting M1 ipRGCs (n = 4).

(D) Analogous to A, but for whole-cell current-clamp recordings from M4 ipRGCs (n = 4).

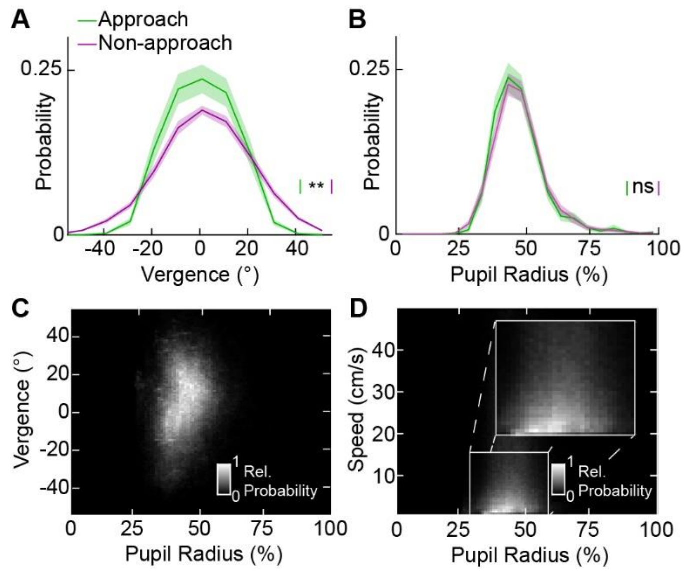

Figure S7. Pupil size, ocular vergence, and speed during cricket hunting. Related to Figure 6.

(A) Probability of ocular vergence values (± SEM) during approach and non-approach periods of the cricket hunting behavior (n = 105 trials, p < 0.01, by bootstrapping). Negative values reflect convergence, while positive values reflect divergence.

(B) Probability of pupil radius values (± SEM) during approach and non-approach periods of the cricket hunting behavior (n = 105, p = 0.71, by bootstrapping) as a percentage of the maximal pupil radius.

(C) Heatmap illustrating the relative probability across all time points of ocular vergence and pupil radius value pairs, weighted equally per mouse (n = 7).

(D) Heatmap illustrating the relative probability across all time points of speed and pupil radius value pairs, weighted equally per mouse (n = 7).

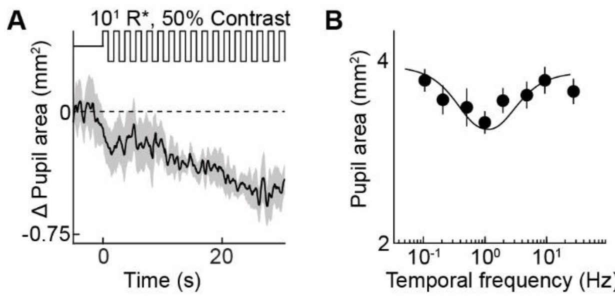

Figure S8. Human pupillary contrast response magnitude. Related to Figure 8.

(A) Average pupillary contrast response trace (± SEM) at $10^{1.5}$ R*, 50% contrast, 1 Hz, as directly measured in mm^2 (n = 7).

(B) Summary data of the pupillary contrast response, as directly measured in mm^2 (n = 7).